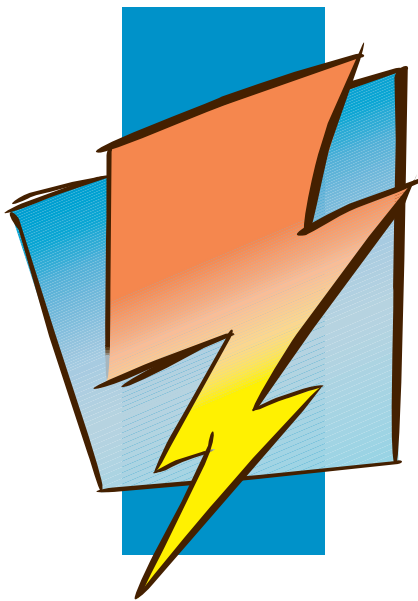


Open Watcom FORTRAN 77

Graphics Library Reference



Version 2.0

Open Watcom

Notice of Copyright

Copyright © 2002-2026 the Open Watcom Contributors. Portions Copyright © 1984-2002 Sybase, Inc. and its subsidiaries. All rights reserved.

Any part of this publication may be reproduced, transmitted, or translated in any form or by any means, electronic, mechanical, manual, optical, or otherwise, without the prior written permission of anyone.

For more information please visit <https://github.com/open-watcom/open-watcom-v2> .

Preface

The Open Watcom FORTRAN 77 Optimizing Compiler is an implementation of the American National Standard programming language FORTRAN, ANSI X3.9-1978, commonly referred to as FORTRAN 77. The language level supported by this compiler includes the full language definition as well as significant extensions to the language.

This manual describes the Open Watcom FORTRAN 77 Graphics Library. This library of routines is used to create graphical images such as lines and circles on the screen. Routines are also provided for displaying text.

This book was produced with the Open Watcom GML electronic publishing system, a software tool developed by WATCOM. In this system, writers use an ASCII text editor to create source files containing text annotated with tags. These tags label the structural elements of the document, such as chapters, sections, paragraphs, and lists. The Open Watcom GML software, which runs on a variety of operating systems, interprets the tags to format the text into a form such as you see here. Writers can produce output for a variety of printers, including laser printers, using separately specified layout directives for such things as font selection, column width and height, number of columns, etc. The result is type-set quality copy containing integrated text and graphics.

Acknowledgements

Many members of Watcom International Corp. have made a significant contribution to the design and implementation of the Open Watcom FORTRAN 77 Graphics Library. The design of this software is based upon ideas evolved and proven over the past decade in other software projects in which these people have been involved.

July, 1997.

Trademarks Used in this Manual

IBM is a registered trademark of International Business Machines Corp.

Hercules is a registered trademark of Hercules Computer Technology.

Microsoft and Windows are registered trademarks of Microsoft Corp.

WATCOM is a trademark of Sybase, Inc. and its subsidiaries.

Table of Contents

1 Graphics Library	1
1.1 Graphics Routines	1
1.2 Graphics Adapters	1
1.3 Classes of Graphics Routines	2
1.3.1 Environment Routines	2
1.3.2 Coordinate System Routines	3
1.3.3 Attribute Routines	4
1.3.4 Drawing Routines	4
1.3.5 Text Routines	5
1.3.6 Graphics Text Routines	5
1.3.7 Image Manipulation Routines	6
1.3.8 Font Manipulation Routines	6
1.3.9 Presentation Graphics Routines	7
1.3.9.1 Display Routines	7
1.3.9.2 Analyze Routines	7
1.3.9.3 Utility Routines	8
1.4 Include Files	8
2 Graphics Library Routines	9
_arc, _arc_w, _arc_wxy	10
_clearscreen	13
_displaycursor	14
_ellipse, _ellipse_w, _ellipse_wxy	15
_floodfill, _floodfill_w	17
_getactivepage	18
_getarcinfo	20
_getbkcolor	22
_getcliprgn	23
_getcolor	24
_getcurrentposition, _getcurrentposition_w	25
_getfillmask	26
_getfontinfo	27
_getgttextextent	29
_getgttextvector	30
_getimage, _getimage_w, _getimage_wxy	31
_getlinestyle	33
_getphyscoord	34
_getpixel, _getpixel_w	35
_getplotaction	36
_gettextcolor	38
_gettextcursor	39
_gettextextent	40
_gettextposition	42
_gettextsettings	43
_gettextwindow	45
_getvideoconfig	46
_getviewcoord, _getviewcoord_w, _getviewcoord_wxy	49
_getvisualpage	51
_getwindowcoord	53
_grstatus	54
_grtext, _grtext_w	55
_imagesize, _imagesize_w, _imagesize_wxy	57

Table of Contents

_lineto, _lineto_w	59
_moveto, _moveto_w	61
_outgtext	62
_outmem	64
_outtext	66
_pg_analyzechart, _pg_analyzechartms	67
_pg_analyzepie	69
_pg_analyzescatter, _pg_analyzescatterms	71
_pg_chart, _pg_chartms	73
_pg_chartpie	76
_pg_chartscatter, _pg_chartscatterms	78
_pg_defaultchart	81
_pg_getchardef	83
_pg_getpalette	85
_pg_getstyleset	88
_pg_hlabelchart	90
_pg_initchart	92
_pg_resetpalette	94
_pg_resetstyleset	97
_pg_setchardef	99
_pg_setpalette	101
_pg_setstyleset	104
_pg_vlabelchart	106
_pie, _pie_w, _pie_wxy	108
_polygon, _polygon_w, _polygon_wxy	111
_putimage, _putimage_w	113
_rectangle, _rectangle_w, _rectangle_wxy	115
_registerfonts	117
_remapallpalette	118
_remappalette	120
_scrolltextwindow	122
_selectpalette	124
_setactivepage	126
_setbkcolor	128
_setcharsize, _setcharsize_w	129
_setcharspacing, _setcharspacing_w	131
_setcliprgn	133
_setcolor	134
_setfillmask	135
_setfont	137
_setgtextvector	139
_setlinestyle	140
_setpixel, _setpixel_w	142
_setplotaction	144
_settextalign	146
_settextcolor	148
_settextcursor	150
_settextorient	152
_settextpath	154
_settextposition	156
_settextrows	157
_settextwindow	159

Table of Contents

_setvideomode	160
_setvideomoderows	163
_setvieworg	164
_setviewport	165
_setvisualpage	166
_setwindow	168
_unregisterfonts	170
_wrapon	171

1 Graphics Library

The Open Watcom FORTRAN 77 Graphics Library consists of a large number of routines that provide graphical image support under DOS and QNX. This chapter provides an overview of this support. The following topics are discussed.

- Graphics Routines
- Graphics Adapters
- Classes of Graphics Routines
 1. Environment Routines
 2. Coordinate System Routines
 3. Attribute Routines
 4. Drawing Routines
 5. Text Routines
 6. Graphics Text Routines
 7. Image Manipulation Routines
 8. Font Manipulation Routines
 9. Presentation Graphics Routines

Display Routines
Analyze Routines
Utility Routines
- Include Files

1.1 Graphics Routines

Graphics routines are used to display graphical images such as lines and circles upon the computer screen. Routines are also provided for displaying text along with the graphics output.

1.2 Graphics Adapters

Support is provided for both color and monochrome screens which are connected to the computer using any of the following graphics adapters:

- IBM Monochrome Display/Printer Adapter (MDPA)
- IBM Color Graphics Adapter (CGA)
- IBM Enhanced Graphics Adapter (EGA)
- IBM Multi-Color Graphics Array (MCGA)

- IBM Video Graphics Array (VGA)
- Hercules Monochrome Adapter
- SuperVGA adapters (SVGA) supplied by various manufacturers

1.3 Classes of Graphics Routines

The routines in the Open Watcom FORTRAN 77 Graphics Library can be organized into a number of classes:

Environment Routines

These routines deal with the hardware environment.

Coordinate System Routines

These routines deal with coordinate systems and mapping coordinates from one system to another.

Attribute Routines

These routines control the display of graphical images.

Drawing Routines

These routines display graphical images such as lines and ellipses.

Text Routines

These routines deal with displaying text in both graphics and text modes.

Graphics Text Routines

These routines deal with displaying graphics text.

Image Manipulation Routines

These routines store and retrieve screen images.

Font Manipulation Routines

These routines deal with displaying font based text.

Presentation Graphics Routines

These routines deal with displaying presentation graphics elements such as bar charts and pie charts.

The following subsections describe these routine classes in more detail. Each routine in the class is noted with a brief description of its purpose.

1.3.1 Environment Routines

These routines deal with the hardware environment. The `_getvideoconfig` routine returns information about the current video mode and the hardware configuration. The `_setvideomode` routine selects a new video mode.

Some video modes support multiple pages of screen memory. The visual page (the one displayed on the screen) may be different than the active page (the one to which objects are being written).

The following routines are defined:

<code>_getactivepage</code>	get the number of the current active graphics page
<code>_getvideoconfig</code>	get information about the graphics configuration
<code>_getvisualpage</code>	get the number of the current visual graphics page
<code>_grstatus</code>	get the status of the most recently called graphics library routine
<code>_setactivepage</code>	set the active graphics page (the page to which graphics objects are drawn)
<code>_settextrows</code>	set the number of rows of text displayed on the screen
<code>_setvideomode</code>	select the video mode to be used
<code>_setvideomoderows</code>	select the video mode and the number of text rows to be used
<code>_setvisualpage</code>	set the visual graphics page (the page displayed on the screen)

1.3.2 Coordinate System Routines

These routines deal with coordinate systems and mapping coordinates from one system to another. The Open Watcom FORTRAN 77 Graphics Library supports three coordinate systems:

1. Physical coordinates
2. View coordinates
3. Window coordinates

Physical coordinates match the physical dimensions of the screen. The physical origin, denoted (0,0), is located at the top left corner of the screen. A pixel to the right of the origin has a positive x-coordinate and a pixel below the origin will have a positive y-coordinate. The x- and y-coordinates will never be negative values.

The view coordinate system can be defined upon the physical coordinate system by moving the origin from the top left corner of the screen to any physical coordinate (see the `_setvieworg` routine). In the view coordinate system, negative x- and y-coordinates are allowed. The scale of the view and physical coordinate systems is identical (both are in terms of pixels)

The window coordinate system is defined in terms of a range of user-specified values (see the `_setwindow` routine). These values are scaled to map onto the physical coordinates of the screen. This allows for consistent pictures regardless of the resolution (number of pixels) of the screen.

The following routines are defined:

<code>_getcliprgn</code>	get the boundary of the current clipping region
<code>_getphyscoord</code>	get the physical coordinates of a point in view coordinates
<code>_getviewcoord</code>	get the view coordinates of a point in physical coordinates
<code>_getviewcoord_w</code>	get the view coordinates of a point in window coordinates
<code>_getviewcoord_wxy</code>	get the view coordinates of a point in window coordinates
<code>_getwindowcoord</code>	get the window coordinates of a point in view coordinates
<code>_setcliprgn</code>	set the boundary of the clipping region
<code>_setvieworg</code>	set the position to be used as the origin of the view coordinate system
<code>_setviewport</code>	set the boundary of the clipping region and the origin of the view coordinate system
<code>_setwindow</code>	define the boundary of the window coordinate system

1.3.3 Attribute Routines

These routines control the display of graphical images such as lines and circles. Lines and figures are drawn using the current color (see the `_setcolor` routine), the current line style (see the `_setlinestyle` routine), the current fill mask (see the `_setfillmask` routine), and the current plotting action (see the `_setplotaction` routine).

The following routines are defined:

<code>_getarcinfo</code>	get the endpoints of the most recently drawn arc
<code>_getbkcolor</code>	get the background color
<code>_getcolor</code>	get the current color
<code>_getfillmask</code>	get the current fill mask
<code>_getlinestyle</code>	get the current line style
<code>_getplotaction</code>	get the current plotting action
<code>_remapallpalette</code>	assign colors for all pixel values
<code>_remappalette</code>	assign color for one pixel value
<code>_selectpalette</code>	select a palette
<code>_setbkcolor</code>	set the background color
<code>_setcolor</code>	set the current color
<code>_setfillmask</code>	set the current fill mask
<code>_setlinestyle</code>	set the current line style
<code>_setplotaction</code>	set the current plotting action

1.3.4 Drawing Routines

These routines display graphical images such as lines and ellipses. Routines exist to draw straight lines (see the `_lineto` routine), rectangles (see the `_rectangle` routine), polygons (see the `_polygon` routine), ellipses (see the `_ellipse` routine), elliptical arcs (see the `_arc` routine) and pie-shaped wedges from ellipses (see the `_pie` routine).

These figures are drawn using the attributes described in the previous section. The routines ending with `_w` or `_wxy` use the window coordinate system; the others use the view coordinate system.

The following routines are defined:

<code>_arc</code>	draw an arc
<code>_arc_w</code>	draw an arc using window coordinates
<code>_arc_wxy</code>	draw an arc using window coordinates
<code>_clearscreen</code>	clear the screen and fill with the background color
<code>_ellipse</code>	draw an ellipse
<code>_ellipse_w</code>	draw an ellipse using window coordinates
<code>_ellipse_wxy</code>	draw an ellipse using window coordinates
<code>_floodfill</code>	fill an area of the screen with the current color
<code>_floodfill_w</code>	fill an area of the screen in window coordinates with the current color
<code>_getcurrentposition</code>	get the coordinates of the current output position
<code>_getcurrentposition_w</code>	get the window coordinates of the current output position
<code>_getpixel</code>	get the color of the pixel at the specified position
<code>_getpixel_w</code>	get the color of the pixel at the specified position in window coordinates
<code>_lineto</code>	draw a line from the current position to a specified position

<code>_lineto_w</code>	draw a line from the current position to a specified position in window coordinates
<code>_moveto</code>	set the current output position
<code>_moveto_w</code>	set the current output position using window coordinates
<code>_pie</code>	draw a wedge of a "pie"
<code>_pie_w</code>	draw a wedge of a "pie" using window coordinates
<code>_pie_wxy</code>	draw a wedge of a "pie" using window coordinates
<code>_polygon</code>	draw a polygon
<code>_polygon_w</code>	draw a polygon using window coordinates
<code>_polygon_wxy</code>	draw a polygon using window coordinates
<code>_rectangle</code>	draw a rectangle
<code>_rectangle_w</code>	draw a rectangle using window coordinates
<code>_rectangle_wxy</code>	draw a rectangle using window coordinates
<code>_setpixel</code>	set the color of the pixel at the specified position
<code>_setpixel_w</code>	set the color of the pixel at the specified position in window coordinates

1.3.5 Text Routines

These routines deal with displaying text in both graphics and text modes. This type of text output can be displayed in only one size.

This text is displayed using the `_outtext` and `_outmem` routines. The output position for text follows the last text that was displayed or can be reset (see the `_settextposition` routine). Text windows can be created (see the `_settextwindow` routine) in which the text will scroll. Text is displayed with the current text color (see the `_settextcolor` routine).

The following routines are defined:

<code>_clearscreen</code>	clear the screen and fill with the background color
<code>_displaycursor</code>	determine whether the cursor is to be displayed after a graphics routine completes execution
<code>_getbkcolor</code>	get the background color
<code>_gettextcolor</code>	get the color used to display text
<code>_gettextcursor</code>	get the shape of the text cursor
<code>_gettextposition</code>	get the current output position for text
<code>_gettextwindow</code>	get the boundary of the current text window
<code>_outmem</code>	display a text string of a specified length
<code>_outtext</code>	display a text string
<code>_scrolltextwindow</code>	scroll the contents of the text window
<code>_setbkcolor</code>	set the background color
<code>_settextcolor</code>	set the color used to display text
<code>_settextcursor</code>	set the shape of the text cursor
<code>_settextposition</code>	set the output position for text
<code>_settextwindow</code>	set the boundary of the region used to display text
<code>_wrapon</code>	permit or disallow wrap-around of text in a text window

1.3.6 Graphics Text Routines

These routines deal with displaying graphics text. Graphics text is displayed as a sequence of line segments, and can be drawn in different sizes (see the `_setcharsize` routine), with different orientations (see the `_settextorient` routine) and alignments (see the `_settextalign` routine). The routines ending with `_w` use the window coordinate system; the others use the view coordinate system.

The following routines are defined:

<code>_gettexttext</code>	get the bounding rectangle for a graphics text string
<code>_gettextsettings</code>	get information about the current settings used to display graphics text
<code>_grtext</code>	display graphics text
<code>_grtext_w</code>	display graphics text using window coordinates
<code>_setcharsize</code>	set the character size used to display graphics text
<code>_setcharsize_w</code>	set the character size in window coordinates used to display graphics text
<code>_setcharspacing</code>	set the character spacing used to display graphics text
<code>_setcharspacing_w</code>	set the character spacing in window coordinates used to display graphics text
<code>_settextalign</code>	set the alignment used to display graphics text
<code>_settextorient</code>	set the orientation used to display graphics text
<code>_settextpath</code>	set the path used to display graphics text

1.3.7 Image Manipulation Routines

These routines are used to transfer screen images. The `_getimage` routine transfers a rectangular image from the screen into memory. The `_putimage` routine transfers an image from memory back onto the screen. The routines ending with `_w` or `_wxy` use the window coordinate system; the others use the view coordinate system.

The following routines are defined:

<code>_getimage</code>	store an image of an area of the screen into memory
<code>_getimage_w</code>	store an image of an area of the screen in window coordinates into memory
<code>_getimage_wxy</code>	store an image of an area of the screen in window coordinates into memory
<code>_imagesize</code>	get the size of a screen area
<code>_imagesize_w</code>	get the size of a screen area in window coordinates
<code>_imagesize_wxy</code>	get the size of a screen area in window coordinates
<code>_putimage</code>	display an image from memory on the screen
<code>_putimage_w</code>	display an image from memory on the screen using window coordinates

1.3.8 Font Manipulation Routines

These routines are for the display of fonts compatible with Microsoft Windows. Fonts are contained in files with an extension of `.FON`. Before font based text can be displayed, the fonts must be registered with the `_registerfonts` routine, and a font must be selected with the `_setfont` routine.

The following routines are defined:

<code>_getfontinfo</code>	get information about the currently selected font
<code>_getgtexttext</code>	get the length in pixels of a text string
<code>_getgtextvector</code>	get the current value of the font text orientation vector
<code>_outgtext</code>	display a string of text in the current font
<code>_registerfonts</code>	initialize the font graphics system
<code>_setfont</code>	select a font from among the registered fonts
<code>_setgtextvector</code>	set the font text orientation vector
<code>_unregisterfonts</code>	free memory allocated by the font graphics system

1.3.9 Presentation Graphics Routines

These routines provide a system for displaying and manipulating presentation graphics elements such as bar charts and pie charts. The presentation graphics routines can be further divided into three classes:

Display Routines

These routines are for the initialization of the presentation graphics system and the displaying of charts.

Analyze Routines

These routines calculate default values for chart elements without actually displaying the chart.

Utility Routines

These routines provide additional support to control the appearance of presentation graphics elements.

The following subsections describe these routine classes in more detail. Each routine in the class is noted with a brief description of its purpose.

1.3.9.1 Display Routines

These routines are for the initialization of the presentation graphics system and the displaying of charts. The `_pg_initchart` routine initializes the system and should be the first presentation graphics routine called. The single-series routines display a single set of data on a chart; the multi-series routines (those ending with `ms`) display several sets of data on the same chart.

The following routines are defined:

<code>_pg_chart</code>	display a bar, column or line chart
<code>_pg_chartms</code>	display a multi-series bar, column or line chart
<code>_pg_chartpie</code>	display a pie chart
<code>_pg_chartscluster</code>	display a scatter chart
<code>_pg_chartsclusterms</code>	display a multi-series scatter chart
<code>_pg_defaultchart</code>	initialize the chart environment for a specific chart type
<code>_pg_initchart</code>	initialize the presentation graphics system

1.3.9.2 Analyze Routines

These routines calculate default values for chart elements without actually displaying the chart. The routines ending with `ms` analyze multi-series charts; the others analyze single-series charts.

The following routines are defined:

<code>_pg_analyzechart</code>	analyze a bar, column or line chart
<code>_pg_analyzechartms</code>	analyze a multi-series bar, column or line chart
<code>_pg_analyzepie</code>	analyze a pie chart
<code>_pg_analyzecluster</code>	analyze a scatter chart
<code>_pg_analyzeclusterms</code>	analyze a multi-series scatter chart

1.3.9.3 Utility Routines

These routines provide additional support to control the appearance of presentation graphics elements.

The following routines are defined:

<code>_pg_getchardef</code>	get bit-map definition for a specific character
<code>_pg_getpalette</code>	get presentation graphics palette (colors, line styles, fill patterns and plot characters)
<code>_pg_getstyleset</code>	get presentation graphics style-set (line styles for window borders and grid lines)
<code>_pg_hlabelchart</code>	display text horizontally on a chart
<code>_pg_resetpalette</code>	reset presentation graphics palette to default values
<code>_pg_resetstyleset</code>	reset presentation graphics style-set to default values
<code>_pg_setchardef</code>	set bit-map definition for a specific character
<code>_pg_setpalette</code>	set presentation graphics palette (colors, line styles, fill patterns and plot characters)
<code>_pg_setstyleset</code>	set presentation graphics style-set (line styles for window borders and grid lines)
<code>_pg_vlabelchart</code>	display text vertically on a chart

1.4 Include Files

All program modules which use the Graphics Library should include the file `graphapi.fi`. This file contains definitions of all the routines in the library. As well, each routine should include `graph.fi` which contains all the structure and constant definitions.

Modules using the presentation graphics routines should also include the file `pgapi.fi`. As well, each routine should include `pg.fi`.

2 Graphics Library Routines

This chapter contains, in alphabetical order, descriptions of the routines which comprise the graphics library. Each description consists of a number of subsections:

Synopsis:

This subsection gives an example of a declaration for the routine, showing the types of the routine and its arguments.

Description:

This subsection is a description of the routine.

Returns:

This subsection describes the return value (if any) for the routine.

See Also:

This subsection provides a list of related routines.

Example:

This subsection consists of an example program demonstrating the use of the routine. In some cases the output from the program is also displayed.

Classification:

This subsection provides an indication of where the routine is commonly found. The following notation is used:

PC Graphics These &routines are part of the PC graphics library.

Systems:

This subsection provides an indication of where the routine is supported. The following notation is used:

DOS This routine is available on both 16-bit DOS and 32-bit extended DOS.

QNX This routine is available on QNX Software Systems' 16 or 32-bit operating systems.

Synopsis:

```
integer*2 function _arc( x1, y1, x2, y2,
                        x3, y3, x4, y4 )

integer*2 x1, y1
integer*2 x2, y2
integer*2 x3, y3
integer*2 x4, y4

integer*2 function _arc_w( x1, y1, x2, y2,
                          x3, y3, x4, y4 )

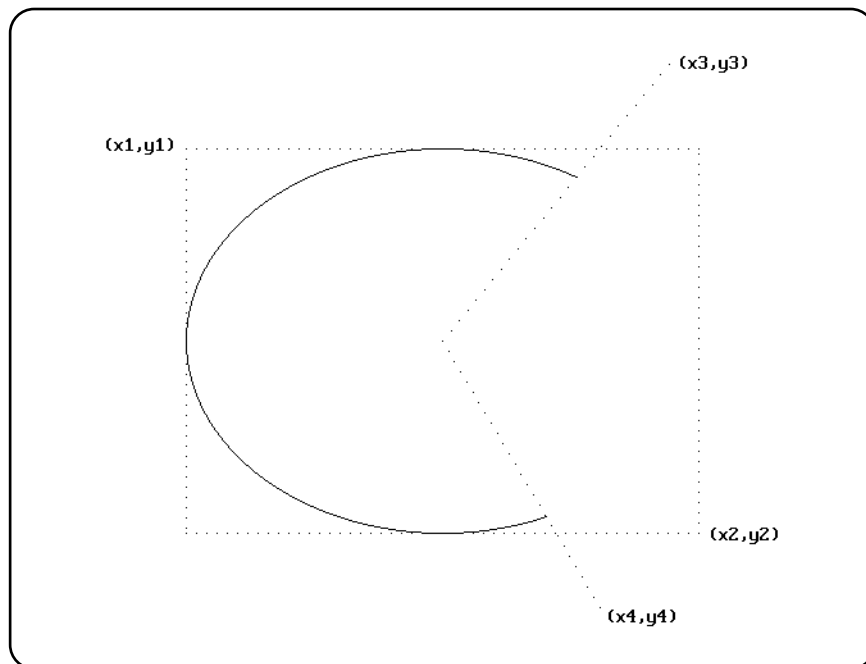
double precision x1, y1
double precision x2, y2
double precision x3, y3
double precision x4, y4

integer*2 function _arc_wxy( p1, p2, p3, p4 )
record /_wxycoord/ p1
record /_wxycoord/ p2
record /_wxycoord/ p3
record /_wxycoord/ p4
```

Description: The `_arc` routines draw elliptical arcs. The `_arc` routine uses the view coordinate system. The `_arc_w` and `_arc_wxy` routines use the window coordinate system.

The center of the arc is the center of the rectangle established by the points $(x1, y1)$ and $(x2, y2)$. The arc is a segment of the ellipse drawn within this bounding rectangle. The arc starts at the point on this ellipse that intersects the vector from the centre of the ellipse to the point $(x3, y3)$. The arc ends at the point on this ellipse that intersects the vector from the centre of the ellipse to the point $(x4, y4)$. The arc is drawn in a counter-clockwise direction with the current plot action using the current color and the current line style.

The following picture illustrates the way in which the bounding rectangle and the vectors specifying the start and end points are defined.



When the coordinates (x1,y1) and (x2,y2) establish a line or a point (this happens when one or more of the x-coordinates or y-coordinates are equal), nothing is drawn.

The current output position for graphics output is set to be the point at the end of the arc that was drawn.

Returns: The `_arc` routines return a non-zero value when the arc was successfully drawn; otherwise, zero is returned.

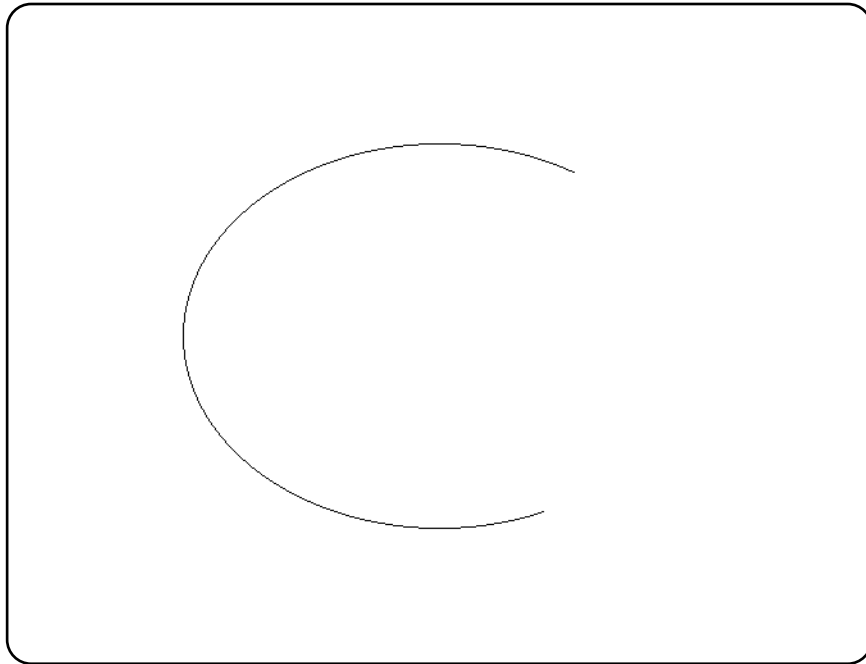
See Also: `_ellipse`, `_pie`, `_rectangle`, `_getarcinfo`, `_setcolor`, `_setlinestyle`, `_setplotaction`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _arc( 120, 90, 520, 390, 500, 20, 450, 460 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: `_arc` - DOS
`_arc_w` - DOS
`_arc_wxy` - DOS

Synopsis: subroutine _clearscreen(area)
 integer*2 area

Description: The _clearscreen routine clears the indicated *area* and fills it with the background color. The *area* argument must be one of the following values:

_GCLEARSCREEN area is entire screen

_GVIEWPORT area is current viewport or clip region

_GWINDOW area is current text window

See Also: _setbkcolor, _setviewport, _setcliprgn, _settextwindow

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _rectangle( _GFILLINTERIOR,
+               100, 100, 540, 380 )
pause
call _setviewport( 200, 200, 440, 280 )
call _clearscreen( _GVIEWPORT )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_displaycursor

Synopsis: integer*2 function _displaycursor(mode)
 integer*2 mode

Description: The _displaycursor routine is used to establish whether the text cursor is to be displayed when graphics routines complete. On entry to a graphics routine, the text cursor is turned off. When the routine completes, the *mode* setting determines whether the cursor is turned back on. The *mode* argument can have one of the following values:

_GCURSORON the cursor will be displayed

_GCURSOROFF the cursor will not be displayed

Returns: The _displaycursor routine returns the previous setting for *mode*.

See Also: _gettextcursor, _settextcursor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

character*30 name

call _setvideomode( _TEXTC80 )
call _settextposition( 2, 1 )
call _displaycursor( _GCURSORON )
call _outtext( 'Cursor ON'//char(10)//char(10)
+             //'Enter your name >'c )
read( *, '(a30)' ) name
call _displaycursor( _GCURSOROFF )
call _settextposition( 6, 1 )
call _outtext( 'Cursor OFF'//char(10)//char(10)
+             //'Enter your name >'c )
read( *, '(a30)' ) name
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis:

```
integer*2 function _ellipse( fill, x1, y1, x2, y2 )
integer*2 fill
integer*2 x1, y1
integer*2 x2, y2

integer*2 function _ellipse_w( fill, x1, y1, x2, y2 )
integer*2 fill,
double precision x1, y1
double precision x2, y2

integer*2 function _ellipse_wxy( fill, p1, p2 )
integer*2 fill,
record /_wxycoord/ p1, p2
```

Description: The `_ellipse` routines draw ellipses. The `_ellipse` routine uses the view coordinate system. The `_ellipse_w` and `_ellipse_wxy` routines use the window coordinate system.

The center of the ellipse is the center of the rectangle established by the points `(x1,y1)` and `(x2,y2)`.

The argument *fill* determines whether the ellipse is filled in or has only its outline drawn. The argument can have one of two values:

`_GFILLINTERIOR` fill the interior by writing pixels with the current plot action using the current color and the current fill mask

`_GBORDER` leave the interior unchanged; draw the outline of the figure with the current plot action using the current color and line style

When the coordinates `(x1,y1)` and `(x2,y2)` establish a line or a point (this happens when one or more of the x-coordinates or y-coordinates are equal), nothing is drawn.

Returns: The `_ellipse` routines return a non-zero value when the ellipse was successfully drawn; otherwise, zero is returned.

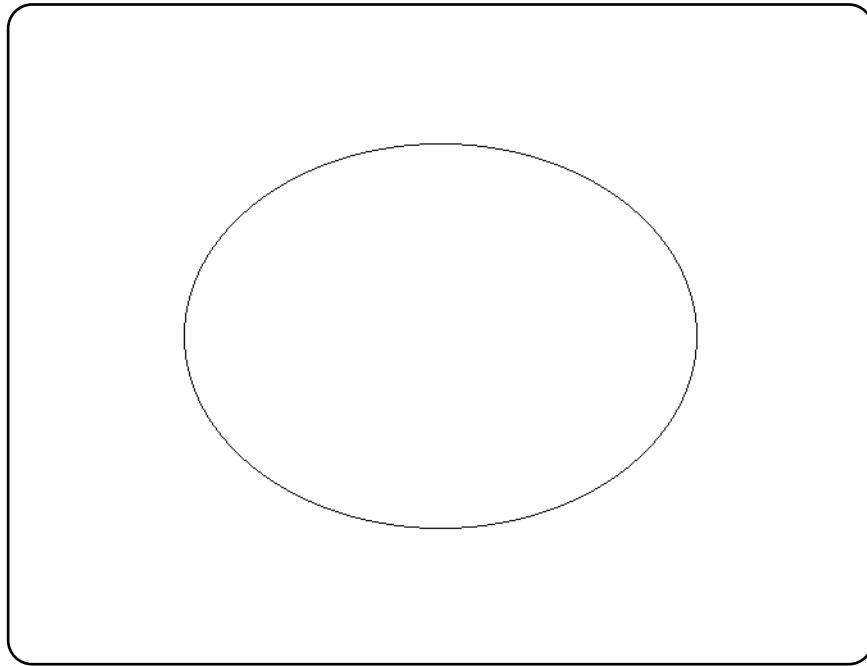
See Also: `_arc`, `_rectangle`, `_setcolor`, `_setfillmask`, `_setlinestyle`, `_setplotaction`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _ellipse( _GBORDER, 120, 90, 520, 390 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: `_ellipse` - DOS
 `_ellipse_w` - DOS
 `_ellipse_wxy` - DOS

Synopsis:

```
integer*2 function _floodfill( x, y, stop_color )
integer*2 x, y
integer*2 stop_color

integer*2 function _floodfill_w( x, y, stop_color )
double precision x, y
integer*2 stop_color
```

Description: The `_floodfill` routines fill an area of the screen. The `_floodfill` routine uses the view coordinate system. The `_floodfill_w` routine uses the window coordinate system.

The filling starts at the point (x, y) and continues in all directions: when a pixel is filled, the neighbouring pixels (horizontally and vertically) are then considered for filling. Filling is done using the current color and fill mask. No filling will occur if the point (x, y) lies outside the clipping region.

If the argument `stop_color` is a valid pixel value, filling will occur in each direction until a pixel is encountered with a pixel value of `stop_color`. The filled area will be the area around (x, y) , bordered by `stop_color`. No filling will occur if the point (x, y) has the pixel value `stop_color`.

If `stop_color` has the value (-1), filling occurs until a pixel is encountered with a pixel value different from the pixel value of the starting point (x, y) . No filling will occur if the pixel value of the point (x, y) is the current color.

Returns: The `_floodfill` routines return zero when no filling takes place; a non-zero value is returned to indicate that filling has occurred.

See Also: `_setcliprpn`, `_setcolor`, `_setfillmask`, `_setplotaction`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _setcolor( 1 )
call _ellipse( _GBORDER, 120, 90, 520, 390 )
call _setcolor( 2 )
call _floodfill( 320, 240, 1 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: `_floodfill` - DOS
`_floodfill_w` - DOS

Synopsis: integer*2 function _getactivepage()

Description: The _getactivepage routine returns the number of the currently selected active graphics page.

Only some combinations of video modes and hardware allow multiple pages of graphics to exist. When multiple pages are supported, the active page may differ from the visual page. The graphics information in the visual page determines what is displayed upon the screen. Animation may be accomplished by alternating the visual page. A graphics page can be constructed without affecting the screen by setting the active page to be different than the visual page.

The number of available video pages can be determined by using the _getvideoconfig routine. The default video page is 0.

Returns: The _getactivepage routine returns the number of the currently selected active graphics page.

See Also: _setactivepage, _setvisualpage, _getvisualpage, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_apage, old_vpage

call _setvideomode( _HRES16COLOR )
old_apage = _getactivepage()
old_vpage = _getvisualpage()
! draw an ellipse on page 0
call _setactivepage( 0 )
call _setvisualpage( 0 )
call _ellipse( _GFILLINTERIOR, 100, 50,
+              540, 150 )
! draw a rectangle on page 1
call _setactivepage( 1 )
call _rectangle( _GFILLINTERIOR, 100, 50,
+               540, 150 )
pause
! display page 1
call _setvisualpage( 1 )
pause
call _setactivepage( old_apage )
call _setvisualpage( old_vpage )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _getarcinfo(start_pt, end_pt,
 inside_pt)

 record /xycoord/ start_pt
 record /xycoord/ end_pt
 record /xycoord/ inside_pt

Description: The _getarcinfo routine returns information about the arc most recently drawn by the _arc or _pie routines. The arguments *start_pt* and *end_pt* are set to contain the endpoints of the arc. The argument *inside_pt* will contain the coordinates of a point within the pie. The points are all specified in the view coordinate system.

The endpoints of the arc can be used to connect other lines to the arc. The interior point can be used to fill the pie.

Returns: The _getarcinfo routine returns a non-zero value when successful. If the previous arc or pie was not successfully drawn, zero is returned.

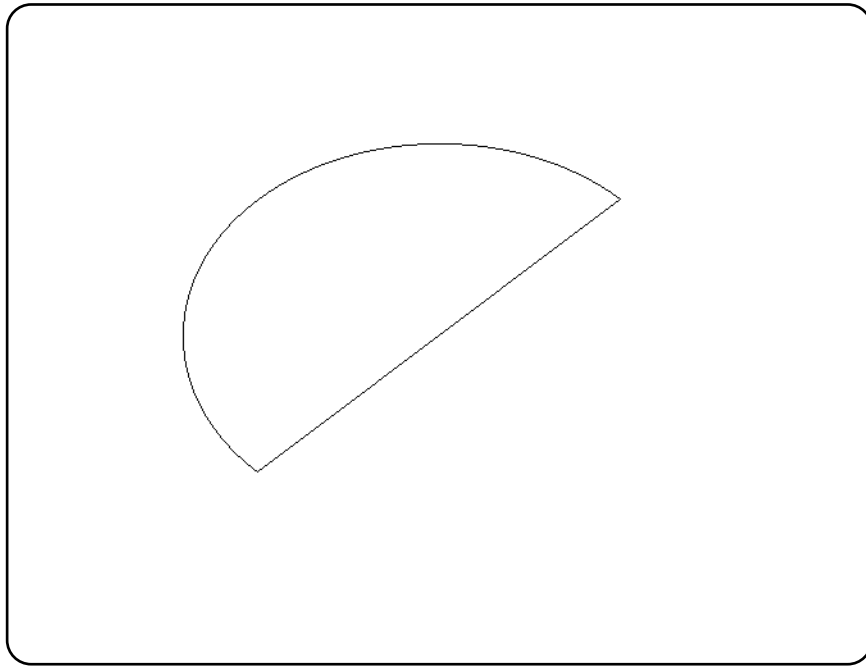
See Also: _arc, _pie

Example: include 'graphapi.fi'
 include 'graph.fi'

 record /xycoord/ start_pt, end_pt, inside_pt

 call _setvideomode(_VRES16COLOR)
 call _arc(120, 90, 520, 390, 520, 90, 120, 390)
 call _getarcinfo(start_pt, end_pt, inside_pt)
 call _moveto(start_pt.xcoord, start_pt.ycoord)
 call _lineto(end_pt.xcoord, end_pt.ycoord)
 pause
 call _setvideomode(_DEFAULTMODE)
 end

produces the following:



Classification: PC Graphics

Systems: DOS

Synopsis: integer*4 function _getbkcolor()

Description: The _getbkcolor routine returns the current background color. In text modes, the background color controls the area behind each individual character. In graphics modes, the background refers to the entire screen. The default background color is 0.

Returns: The _getbkcolor routine returns the current background color.

See Also: _setbkcolor, _remappalette

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer bk, old_bk
integer colors(16)/
+   _BLACK, _BLUE, _GREEN,
+   _CYAN, _RED, _MAGENTA,
+   _BROWN, _WHITE, _GRAY, _LIGHTBLUE,
+   _LIGHTGREEN, _LIGHTCYAN, _LIGHTRED,
+   _LIGHTMAGENTA, _YELLOW, _BRIGHTWHITE/

call _setvideomode( _VRES16COLOR )
old_bk = _getbkcolor()
do bk = 1, 16
    call _setbkcolor( colors( bk ) )
    pause
enddo
call _setbkcolor( old_bk )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _getcliprpn(x1, y1, x2, y2)
 integer*2 x1, y1
 integer*2 x2, y2

Description: The _getcliprpn routine returns the location of the current clipping region. A clipping region is defined with the _setcliprpn or _setviewport routines. By default, the clipping region is the entire screen.

The current clipping region is a rectangular area of the screen to which graphics output is restricted. The top left corner of the clipping region is placed in the arguments (x1,y1) . The bottom right corner of the clipping region is placed in (x2,y2) .

See Also: _setcliprpn, _setviewport

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*2 x1, y1, x2, y2

call _setvideomode( _VRES16COLOR )
call _getcliprpn( x1, y1, x2, y2 )
call _setcliprpn( 130, 100, 510, 380 )
call _ellipse( _GBORDER, 120, 90, 520, 390 )
pause
call _setcliprpn( x1, y1, x2, y2 )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_getcolor

Synopsis: integer*2 function _getcolor()

Description: The _getcolor routine returns the pixel value for the current color. This is the color used for displaying graphics output. The default color value is one less than the maximum number of colors in the current video mode.

Returns: The _getcolor routine returns the pixel value for the current color.

See Also: _setcolor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer col, old_col

call _setvideomode( _VRES16COLOR )
old_col = _getcolor()
do col = 0, 15
    call _setcolor( col )
    call _rectangle( _GFILLINTERIOR,
+                  100, 100, 540, 380 )
    pause
enddo
call _setcolor( old_col )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: record /xycoord/ function _getcurrentposition()
 record /_wxycoord/ function _getcurrentposition_w()

Description: The _getcurrentposition routines return the current output position for graphics. The _getcurrentposition routine returns the point in view coordinates. The _getcurrentposition_w routine returns the point in window coordinates.

The current position defaults to the origin, (0,0), when a new video mode is selected. It is changed by successful calls to the _arc, _moveto and _lineto routines as well as the _setviewport routine.

Note that the output position for graphics output differs from that for text output. The output position for text output can be set by use of the _settextposition routine.

Returns: The _getcurrentposition routines return the current output position for graphics.

See Also: _moveto, _settextposition

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ old_pos

call _setvideomode( _VRES16COLOR )
old_pos = _getcurrentposition()
call _moveto( 100, 100 )
call _lineto( 540, 100 )
call _lineto( 320, 380 )
call _lineto( 100, 100 )
call _moveto( old_pos.xcoord, old_pos.ycoord )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _getcurrentposition - DOS
 _getcurrentposition_w - DOS

Synopsis: subroutine _getfillmask(mask)
 integer*1 mask(8)

Description: The _getfillmask routine copies the current fill mask into the area located by the argument *mask*. The fill mask is used by the _ellipse, _floodfill, _pie, _polygon and _rectangle routines that fill an area of the screen.

The fill mask is an eight-byte array which is interpreted as a square pattern (8 by 8) of 64 bits. Each bit in the mask corresponds to a pixel. When a region is filled, each point in the region is mapped onto the fill mask. When a bit from the mask is one, the pixel value of the corresponding point is set using the current plotting action with the current color; when the bit is zero, the pixel value of that point is not affected.

When the fill mask is not set, a fill operation will set all points in the fill region to have a pixel value of the current color.

See Also: _floodfill, _setfillmask, _setplotaction

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*1 old_mask(8)
integer*1 new_mask(8)/
+           '81'x, '42'x, '24'x, '18'x,
+           '18'x, '24'x, '42'x, '81'x/

call _setvideomode( _VRES16COLOR )
call _getfillmask( old_mask )
call _setfillmask( new_mask )
call _rectangle( _GFillInterior,
+               100, 100, 540, 380 )
call _setfillmask( old_mask )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _getfontinfo(info)
 record /_fontinfo/ info

Description: The _getfontinfo routine returns information about the currently selected font. Fonts are selected with the _setFont routine. The font information is returned in the _fontinfo structure indicated by the argument *info*. The structure contains the following fields:

<i>type</i>	1 for a vector font, 0 for a bit-mapped font
<i>ascent</i>	distance from top of character to baseline in pixels
<i>pixwidth</i>	character width in pixels (0 for a proportional font)
<i>pixheight</i>	character height in pixels
<i>avgwidth</i>	average character width in pixels
<i>filename</i>	name of the file containing the current font
<i>facename</i>	name of the current font

Returns: The _getfontinfo routine returns zero if the font information is returned successfully; otherwise a negative value is returned.

See Also: _registerfonts, _unregisterfonts, _setFont, _outgtext, _getgtextextent,
 _setgtextvector, _getgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /_fontinfo/ info

integer width

call _setvideomode( _VRES16COLOR )
call _getfontinfo( info )
call _moveto( 100, 100 )
call _outgtext( 'WATCOM Graphics'c )
width = _getgtextextent( 'WATCOM Graphics'c )
call _rectangle( _GBORDER, 100, 100,
+               100 + width, 100 + info.pixheight )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function __getgtextextent(text)
 character*(*) text

Description: The __getgtextextent routine returns the length in pixels of the argument *text* as it would be displayed in the current font by the routine __outgtext. Note that the text is not displayed on the screen, only its length is determined.

Returns: The __getgtextextent routine returns the length in pixels of a string.

See Also: __registerfonts, __unregisterfonts, __setfont, __getfontinfo, __outgtext,
 __setgtextvector, __getgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /__fontinfo/ info

integer width

call __setvideomode( __VRES16COLOR )
call __getfontinfo( info )
call __moveto( 100, 100 )
call __outgtext( 'WATCOM Graphics'c )
width = __getgtextextent( 'WATCOM Graphics'c )
call __rectangle( __GBORDER, 100, 100,
+               100 + width, 100 + info.pixheight )
pause
call __setvideomode( __DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_getgtextvector

Synopsis: record /xycoord/ function _getgtextvector()

Description: The _getgtextvector routine returns the current value of the text orientation vector. This is the direction used when text is displayed by the _outgtext routine.

Returns: The _getgtextvector routine returns, as an xycoord structure, the current value of the text orientation vector.

See Also: _registerfonts, _unregisterfonts, _setfont, _getfontinfo, _outgtext, _getgtexttextent, _setgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ old_vec

call _setvideomode( _VRES16COLOR )
old_vec = _getgtextvector()
call _setgtextvector( 0, -1 )
call _moveto( 100, 100 )
call _outgtext( 'WATCOM Graphics'c )
call _setgtextvector( old_vec.xcoord, old_vec.ycoord )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis:

```
subroutine _getimage( x1, y1, x2, y2, image )
integer*2 x1, y1
integer*2 x2, y2
integer*1 image(*)

subroutine _getimage_w( x1, y1, x2, y2, image )
double precision x1, y1
double precision x2, y2
integer*1 image(*)

subroutine _getimage_wxy( p1, p2, image )
record /_wxycoord/ p1, p2
integer*1 image(*)
```

Description: The `_getimage` routines store a copy of an area of the screen into the buffer indicated by the *image* argument. The `_getimage` routine uses the view coordinate system. The `_getimage_w` and `_getimage_wxy` routines use the window coordinate system.

The screen image is the rectangular area defined by the points $(x1, y1)$ and $(x2, y2)$. The buffer *image* must be large enough to contain the image (the size of the image can be determined by using the `_imagesize` routine). The image may be displayed upon the screen at some later time by using the `_putimage` routines.

See Also: `_imagesize`, `_putimage`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*1 image(:)
integer y, image_size, istat

call _setvideomode( _VRES16COLOR )
call _ellipse( _GFILLINTERIOR,
+             100, 100, 200, 200 )
image_size = _imagesize( 100, 100, 201, 201 )
allocate( image(image_size), stat = istat )
if( istat .eq. 0 )then
    call _getimage( 100, 100, 201, 201, image )
    call _putimage( 260, 200, image, _GPSET )
    call _putimage( 420, 100, image, _GPSET )
    do y = 100, 280, 20
        call _putimage( 420, y, image, _GXOR )
        call _putimage( 420, y+20, image, _GXOR )
    enddo
    deallocate( image )
endif
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: `_getimage` - DOS
`_getimage_w` - DOS
`_getimage_wxy` - DOS

Synopsis: integer*2 function _getlinestyle()

Description: The _getlinestyle routine returns the current line-style mask.

The line-style mask determines the style by which lines and arcs are drawn. The mask is treated as an array of 16 bits. As a line is drawn, a pixel at a time, the bits in this array are cyclically tested. When a bit in the array is 1, the pixel value for the current point is set using the current color according to the current plotting action; otherwise, the pixel value for the point is left unchanged. A solid line would result from a value of 'FFFF'x and a dashed line would result from a value of 'F0F0'x.

The default line style mask is 'FFFF'x.

Returns: The _getlinestyle routine returns the current line-style mask.

See Also: _lineto, _pie, _rectangle, _polygon, _setlinestyle

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer DASHED
parameter (DASHED='f0f0'x)

integer old_style

call _setvideomode( _VRES16COLOR )
old_style = _getlinestyle()
call _setlinestyle( DASHED )
call _rectangle( _GBORDER, 100, 100, 540, 380 )
call _setlinestyle( old_style )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_getphyscoord

Synopsis: record /xycoord/ function _getphyscoord(x, y)
 integer*2 x, y

Description: The _getphyscoord routine returns the physical coordinates of the position with view coordinates (x,y) . View coordinates are defined by the _setvieworg and _setviewport routines.

Returns: The _getphyscoord routine returns the physical coordinates, as an xycoord structure, of the given point.

See Also: _getviewcoord, _setvieworg, _setviewport

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ pos

real urand
integer seed

seed = 75347
call _setvideomode( _VRES16COLOR )
call _setvieworg(
+      mod( int( urand( seed )*32767 ), 640 ),
+      mod( int( urand( seed )*32767 ), 480 ) )
pos = _getphyscoord( 0, 0 )
call _rectangle( _GBORDER,
+      - pos.xcoord, - pos.ycoord,
+      639 - pos.xcoord, 479 - pos.ycoord )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _getpixel(x, y)
 integer*2 x, y

 integer*2 function _getpixel_w(x, y)
 double precision x, y

Description: The _getpixel routines return the pixel value for the point with coordinates (x,y) . The _getpixel routine uses the view coordinate system. The _getpixel_w routine uses the window coordinate system.

Returns: The _getpixel routines return the pixel value for the given point when the point lies within the clipping region; otherwise, (-1) is returned.

See Also: _setpixel

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer x, y, i
real urand
integer seed

seed = 75347
call _setvideomode( _VRES16COLOR )
call _rectangle( _GBORDER, 100, 100, 540, 380 )
do i = 0, 60000
  x = 101 + mod( int( urand( seed )*32767 ),
+               439 )
  y = 101 + mod( int( urand( seed )*32767 ),
+               279 )
  call _setcolor( _getpixel( x, y ) + 1 )
  call _setpixel( x, y )
enddo
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _getpixel - DOS
 _getpixel_w - DOS

Synopsis: integer*2 function _getplotaction()

Description: The _getplotaction routine returns the current plotting action.

The drawing routines cause pixels to be set with a pixel value. By default, the value to be set is obtained by replacing the original pixel value with the supplied pixel value. Alternatively, the replaced value may be computed as a function of the original and the supplied pixel values.

The plotting action can have one of the following values:

<u>_GPSET</u>	replace the original screen pixel value with the supplied pixel value
<u>_GAND</u>	replace the original screen pixel value with the <i>bitwise and</i> of the original pixel value and the supplied pixel value
<u>_GOR</u>	replace the original screen pixel value with the <i>bitwise or</i> of the original pixel value and the supplied pixel value
<u>_GXOR</u>	replace the original screen pixel value with the <i>bitwise exclusive-or</i> of the original pixel value and the supplied pixel value. Performing this operation twice will restore the original screen contents, providing an efficient method to produce animated effects.

Returns: The _getplotaction routine returns the current plotting action.

See Also: _setplotaction

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_act

call _setvideomode( _VRES16COLOR )
old_act = _getplotaction()
call _setplotaction( _GPSET )
call _rectangle( _GFILLINTERIOR, 100, 100,
+               540, 380 )
pause
call _setplotaction( _GXOR )
call _rectangle( _GFILLINTERIOR, 100, 100,
+               540, 380 )
pause
call _setplotaction( old_act )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_gettextcolor

Synopsis: integer*2 function _gettextcolor()

Description: The _gettextcolor routine returns the pixel value of the current text color. This is the color used for displaying text with the _outtext and _outmem routines. The default text color value is set to 7 whenever a new video mode is selected.

Returns: The _gettextcolor routine returns the pixel value of the current text color.

See Also: _settextcolor, _setcolor, _outtext, _outmem

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_col
integer old_bk

call _setvideomode( _TEXTC80 )
old_col = _gettextcolor()
old_bk = _getbkcolor()
call _settextcolor( 7 )
call _setbkcolor( _BLUE )
call _outtext( ' WATCOM '//char(10)//
+           'Graphics'c )
call _settextcolor( old_col )
call _setbkcolor( old_bk )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function __gettextcursor()

Description: The __gettextcursor routine returns the current cursor attribute, or shape. The cursor shape is set with the __settextcursor routine. See the __settextcursor routine for a description of the value returned by the __gettextcursor routine.

Returns: The __gettextcursor routine returns the current cursor shape when successful; otherwise, (-1) is returned.

See Also: __settextcursor, __displaycursor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*2 old_shape

old_shape = __gettextcursor()
call __settextcursor( '0007'x )
call __outtext(
+   char(10)//'Block cursor'c )
pause
call __settextcursor( '0407'x )
call __outtext(
+   char(10)//'Half height cursor'c )
pause
call __settextcursor( '2000'x )
call __outtext(
+   char(10)//'No cursor'c )
pause
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _gettextextent(x, y, text, concat, extent)
 integer*2 x, y
 character*(*) text
 record /xycoord/ concat
 record /xycoord/ extent(4)

Description: The _gettextextent routine simulates the effect of using the _grtext routine to display the text string *text* at the position (x,y), using the current text settings. The concatenation point is returned in the argument *concat*. The text extent parallelogram is returned in the array *extent*.

The concatenation point is the position to use to output text after the given string. The text extent parallelogram outlines the area where the text string would be displayed. The four points are returned in counter-clockwise order, starting at the upper-left corner.

See Also: _grtext, _gettextsettings

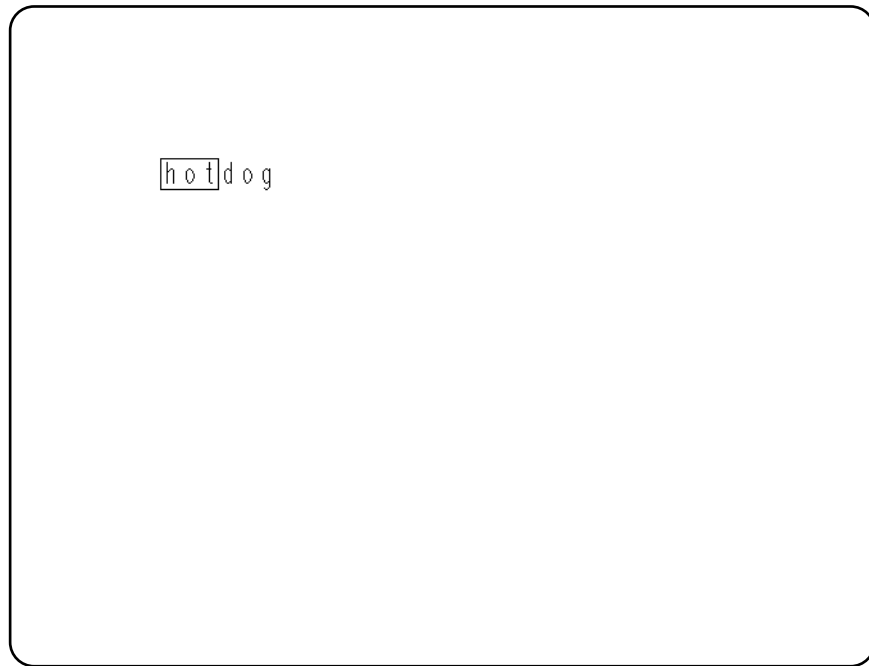
Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ concat, extent(4)

call _setvideomode( _VRES16COLOR )
call _grtext( 100, 100, 'hot'c )
call _gettextextent( 100, 100, 'hot'c,
+                   concat, extent )
call _polygon( _GBORDER, 4, extent )
call _grtext( concat.xcoord, concat.ycoord,
+           'dog'c )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

_gettextposition

Synopsis: record /rccoord/ function _gettextposition()

Description: The _gettextposition routine returns the current output position for text. This position is in terms of characters, not pixels.

The current position defaults to the top left corner of the screen, (1,1), when a new video mode is selected. It is changed by successful calls to the _outtext, _outmem, _settextposition and _settextwindow routines.

Note that the output position for graphics output differs from that for text output. The output position for graphics output can be set by use of the _moveto routine.

Returns: The _gettextposition routine returns, as an rccoord structure, the current output position for text.

See Also: _outtext, _outmem, _settextposition, _settextwindow, _moveto

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /rccoord/ old_pos

call _setvideomode( _TEXTC80 )
old_pos = _gettextposition()
call _settextposition( 10, 40 )
call _outtext( 'WATCOM Graphics'c )
call _settextposition( old_pos.row, old_pos.col )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _gettextsettings(settings)
 record /textsettings/ settings

Description: The _gettextsettings routine returns information about the current text settings used when text is displayed by the _grtext routine. The information is stored in the textsettings structure indicated by the argument *settings*. The structure contains the following fields (all are integer*2 fields):

<i>basevectorx</i>	x-component of the current base vector
<i>basevectory</i>	y-component of the current base vector
<i>path</i>	current text path
<i>height</i>	current text height (in pixels)
<i>width</i>	current text width (in pixels)
<i>spacing</i>	current text spacing (in pixels)
<i>horizontaln</i>	horizontal component of the current text alignment
<i>verticaln</i>	vertical component of the current text alignment

See Also: _grtext, _setcharsize, _setcharspacing, _setttextalign, _setttextpath,
 _setttextorient

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /textsettings/ ts

call _setvideomode( _VRES16COLOR )
call _gettextsettings( ts )
call _grtext( 100, 100, 'WATCOM'c )
call _setcharsize( 2 * ts.height, 2 * ts.width )
call _grtext( 100, 300, 'Graphics'c )
call _setcharsize( ts.height, ts.width )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _gettextwindow(row1, col1, row2, col2)
 integer*2 row1, col1
 integer*2 row2, col2

Description: The _gettextwindow routine returns the location of the current text window. A text window is defined with the _settextwindow routine. By default, the text window is the entire screen.

The current text window is a rectangular area of the screen. Text display is restricted to be within this window. The top left corner of the text window is placed in the arguments (row1,col1) . The bottom right corner of the text window is placed in (row2,col2) .

See Also: _settextwindow, _outtext, _outmem, _settextposition, _scrolltextwindow

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i
integer*2 r1, c1, r2, c2
character*80 buff

call _setvideomode( _TEXTC80 )
call _gettextwindow( r1, c1, r2, c2 )
call _settextwindow( 5, 20, 20, 40 )
do i = 1, 20
    write( buff, '( "Line "', i2, a1, a1 )' )
+      i, char(10), char(0)
    call _outtext( buff )
enddo
pause
call _settextwindow( r1, c1, r2, c2 )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine `_getvideoconfig( config )`
 record `/videoconfig/ config`

Description: The `_getvideoconfig` routine returns information about the current video mode and the hardware configuration. The information is returned in the `videoconfig` structure indicated by the argument `config`. The structure contains the following fields (all are `integer*2` fields):

<i>numxpixels</i>	number of pixels in x-axis
<i>numypixels</i>	number of pixels in y-axis
<i>numtextcols</i>	number of text columns
<i>numtextrows</i>	number of text rows
<i>numcolors</i>	number of actual colors
<i>bitsperpixel</i>	number of bits in a pixel value
<i>numvideopages</i>	number of video pages
<i>mode</i>	current video mode
<i>adapter</i>	adapter type
<i>monitor</i>	monitor type
<i>memory</i>	number of kilobytes (1024 characters) of video memory

The `adapter` field will contain one of the following values:

<i><u>_NODISPLAY</u></i>	no display adapter attached
<i><u>_UNKNOWN</u></i>	unknown adapter/monitor type
<i><u>_MDPA</u></i>	Monochrome Display/Printer Adapter
<i><u>_CGA</u></i>	Color Graphics Adapter
<i><u>_HERCULES</u></i>	Hercules Monochrome Adapter
<i><u>_MCGA</u></i>	Multi-Color Graphics Array
<i><u>_EGA</u></i>	Enhanced Graphics Adapter
<i><u>_VGA</u></i>	Video Graphics Array
<i><u>_SVGA</u></i>	SuperVGA Adapter

The `monitor` field will contain one of the following values:

<i>__MONO</i>	regular monochrome
<i>__COLOR</i>	regular color
<i>__ENHANCED</i>	enhanced color
<i>__ANALOGMONO</i>	analog monochrome
<i>__ANALOGCOLOR</i>	analog color

The amount of memory reported by `__getvideoconfig` will not always be correct for SuperVGA adapters. Since it is not always possible to determine the amount of memory, `__getvideoconfig` will always report 256K, the minimum amount.

See Also: `__setvideomode`, `__setvideomoderows`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer mode
record /videoconfig/ vc
character*80 buff

call _getvideoconfig( vc )
select( vc.adapter )
case( _VGA, _SVGA )
    mode = _VRES16COLOR
case( _MCGA )
    mode = _MRES256COLOR
case( _EGA )
    if( vc.monitor .eq. _MONO )then
        mode = _ERESNOCOLOR
    else
        mode = _ERESCOLOR
    endif
case( _CGA )
    mode = _MRES4COLOR
case( _HERCULES )
    mode = _HERCMONO
case default
    stop 'No graphics adapter'
endselect
if( _setvideomode( mode ) .ne. 0 )then
    call _getvideoconfig( vc )
    write( buff,
+         '( i3, '' x '', i3, '' x '', i3, a1 )' )
+         vc.numxpixels, vc.numypixels,
+         vc.numcolors, char(0)
    call _outtext( buff )
    pause
    call _setvideomode( _DEFAULTMODE )
endif
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: record /xycoord/ function _getviewcoord(x, y)
 integer*2 x, y

 record /xycoord/ function _getviewcoord_w(x, y)
 double precision x, y

 record /xycoord/ function _getviewcoord_wxy(p)
 record /_wxycoord/ p

Description: The _getviewcoord routines translate a point from one coordinate system to viewport coordinates. The _getviewcoord routine translates the point (x, y) from physical coordinates. The _getviewcoord_w and _getviewcoord_wxy routines translate the point from the window coordinate system.

Viewport coordinates are defined by the _setvieworg and _setviewport routines. Window coordinates are defined by the _setwindow routine.

Returns: The _getviewcoord routines return the viewport coordinates, as an xycoord structure, of the given point.

See Also: _getphyscoord, _setvieworg, _setviewport, _setwindow

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ pos1, pos2

integer seed
real urand

seed = 75347
call _setvideomode( _VRES16COLOR )
call _setvieworg(
+      mod( int( urand( seed )*32767 ), 640 ),
+      mod( int( urand( seed )*32767 ), 480 ) )
pos1 = _getviewcoord( 0, 0 )
pos2 = _getviewcoord( 639, 479 )
call _rectangle( _GBORDER,
+               pos1.xcoord, pos1.ycoord,
+               pos2.xcoord, pos2.ycoord )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _getviewcoord - DOS
 _getviewcoord_w - DOS
 _getviewcoord_wxy - DOS

Synopsis: integer*2 function _getvisualpage()

Description: The _getvisualpage routine returns the number of the currently selected visual graphics page.

Only some combinations of video modes and hardware allow multiple pages of graphics to exist. When multiple pages are supported, the active page may differ from the visual page. The graphics information in the visual page determines what is displayed upon the screen. Animation may be accomplished by alternating the visual page. A graphics page can be constructed without affecting the screen by setting the active page to be different than the visual page.

The number of available video pages can be determined by using the _getvideoconfig routine. The default video page is 0.

Returns: The _getvisualpage routine returns the number of the currently selected visual graphics page.

See Also: _setvisualpage, _setactivepage, _getactivepage, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_apage, old_vpage

call _setvideomode( _HRES16COLOR )
old_apage = _getactivepage()
old_vpage = _getvisualpage()
! draw an ellipse on page 0
call _setactivepage( 0 )
call _setvisualpage( 0 )
call _ellipse( _GFILLINTERIOR, 100, 50,
+              540, 150 )
! draw a rectangle on page 1
call _setactivepage( 1 )
call _rectangle( _GFILLINTERIOR, 100, 50,
+               540, 150 )
pause
! display page 1
call _setvisualpage( 1 )
pause
call _setactivepage( old_apage )
call _setvisualpage( old_vpage )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: record /_wxycoord/ function _getwindowcoord(x, y)
 integer*2 x, y

Description: The _getwindowcoord routine returns the window coordinates of the position with view coordinates (x,y) . Window coordinates are defined by the _setwindow routine.

Returns: The _getwindowcoord routine returns the window coordinates, as a _wxycoord structure, of the given point.

See Also: _setwindow, _getviewcoord

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ centre
record /_wxycoord/ pos1, pos2

call _setvideomode( _MAXRESMODE )
! draw a box 50 pixels square
! in the middle of the screen
centre = _getviewcoord_w( 0.5, 0.5 )
pos1 = _getwindowcoord( centre.xcoord - 25,
+                       centre.ycoord - 25 )
pos2 = _getwindowcoord( centre.xcoord + 25,
+                       centre.ycoord + 25 )
call _rectangle_wxy( _GBORDER, pos1, pos2 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _grstatus()

Description: The _grstatus routine returns the status of the most recently called graphics library routine. The routine can be called after any graphics routine to determine if any errors or warnings occurred. The routine returns 0 if the previous routine was successful. Values less than 0 indicate an error occurred; values greater than 0 indicate a warning condition.

The following values can be returned:

Constant	Value	Explanation
_GROK	0	no error
_GRError	-1	graphics error
_GRMODENOTSUPPORTED	-2	video mode not supported
_GRNOTINPROPERMODE	-3	routine n/a in this mode
_GRINVALIDPARAMETER	-4	invalid parameter(s)
_GRINSUFFICIENTMEMORY	-5	out of memory
_GRFONTFILENOTFOUND	-6	can't open font file
_GRINVALIDFONTFILE	-7	font file has invalid format
_GRNOOUTPUT	1	nothing was done
_GRCLIPPED	2	output clipped

Returns: The _grstatus routine returns the status of the most recently called graphics library routine.

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer x, y
real urand
integer seed

seed = 75347
call _setvideomode( _VRES16COLOR )
while( _grstatus() .eq. _GROK )do
    x = mod( int( urand( seed )*32767 ), 700 )
    y = mod( int( urand( seed )*32767 ), 500 )
    call _setpixel( x, y )
endwhile
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis:

```
integer*2 function __grtext( x, y, text )
integer*2 x, y
character*(*) text

integer*2 function __grtext_w( x, y, text )
double precision x, y
character*(*) text
```

Description: The `__grtext` routines display a character string. The `__grtext` routine uses the view coordinate system. The `__grtext_w` routine uses the window coordinate system.

The character string *text* is displayed at the point *(x,y)* . The string must be terminated by a null character (`char(0)`). The text is displayed in the current color using the current text settings.

The graphics library can display text in three different ways.

1. The `__outtext` and `__outmem` routines can be used in any video mode. However, this variety of text can be displayed in only one size.
2. The `__grtext` routine displays text as a sequence of line segments, and can be drawn in different sizes, with different orientations and alignments.
3. The `__outgtext` routine displays text in the currently selected font. Both bit-mapped and vector fonts are supported; the size and type of text depends on the fonts that are available.

Returns: The `__grtext` routines return a non-zero value when the text was successfully drawn; otherwise, zero is returned.

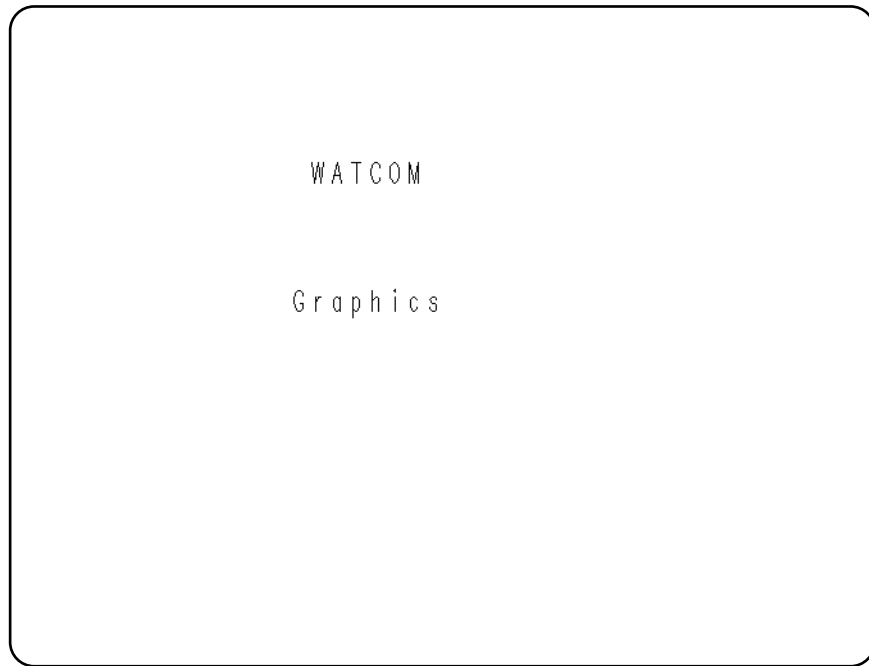
See Also: `__outtext`, `__outmem`, `__outgtext`, `__setcharsize`, `__setttextalign`, `__setttextpath`, `__setttextorient`, `__setcharspacing`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call __setvideomode( _VRES16COLOR )
call __grtext( 200, 100, ' WATCOM'c )
call __grtext( 200, 200, 'Graphics'c )
pause
call __setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: _grtext - DOS
 _grtext_w - DOS

Synopsis:

```
integer*4 function _imagesize( x1, y1, x2, y2 )
integer*2 x1, y1
integer*2 x2, y2

integer*4 function _imagesize_w( x1, y1, x2, y2 )
double precision x1, y1
double precision x2, y2

integer*4 function _imagesize_wxy( p1, p2 )
record /_wxycoord/ p1, p2
```

Description: The `_imagesize` routines compute the number of bytes required to store a screen image. The `_imagesize` routine uses the view coordinate system. The `_imagesize_w` and `_imagesize_wxy` routines use the window coordinate system.

The screen image is the rectangular area defined by the points $(x1, y1)$ and $(x2, y2)$. The storage area used by the `_getimage` routines must be at least this large (in bytes).

Returns: The `_imagesize` routines return the size of a screen image.

See Also: `_getimage`, `_putimage`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*1 image(:)
integer y, image_size, istat

call _setvideomode( _VRES16COLOR )
call _ellipse( _GFILLINTERIOR,
+             100, 100, 200, 200 )
image_size = _imagesize( 100, 100, 201, 201 )
allocate( image(image_size), stat = istat )
if( istat .eq. 0 )then
    call _getimage( 100, 100, 201, 201, image )
    call _putimage( 260, 200, image, _GPSET )
    call _putimage( 420, 100, image, _GPSET )
    do y = 100, 280, 20
        call _putimage( 420, y, image, _GXOR )
        call _putimage( 420, y+20, image, _GXOR )
    enddo
    deallocate( image )
endif
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: `_imagesize` - DOS
 `_imagesize_w` - DOS
 `_imagesize_wxy` - DOS

Synopsis: integer*2 function _lineto(x, y)
 integer*2 x, y

integer*2 function _lineto_w(x, y)
double precision x, y

Description: The _lineto routines draw straight lines. The _lineto routine uses the view coordinate system. The _lineto_w routine uses the window coordinate system.

The line is drawn from the current position to the point at the coordinates (x,y) . The point (x,y) becomes the new current position. The line is drawn with the current plotting action using the current line style and the current color.

Returns: The _lineto routines return a non-zero value when the line was successfully drawn; otherwise, zero is returned.

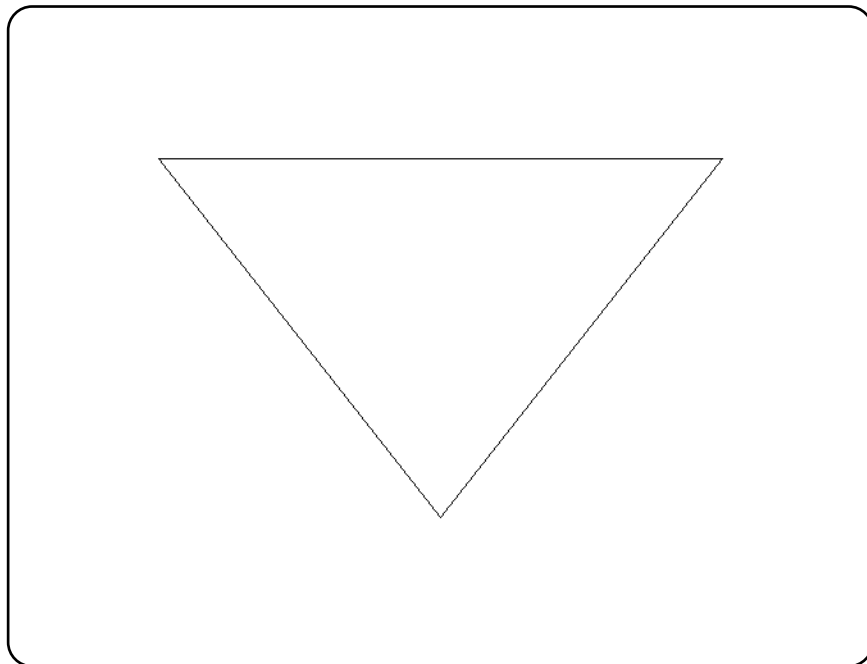
See Also: _moveto, _setcolor, _setlinestyle, _setplotaction

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _moveto( 100, 100 )
call _lineto( 540, 100 )
call _lineto( 320, 380 )
call _lineto( 100, 100 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



_lineto Routines

Classification: PC Graphics

Systems: `_lineto` - DOS
 `_lineto_w` - DOS

Synopsis: record /xycoord/ function _moveto(x, y)
 integer*2 x, y

 record /wxycoord/ function _moveto_w(x, y)
 double precision x, y

Description: The _moveto routines set the current output position for graphics. The _moveto routine uses the view coordinate system. The _moveto_w routine uses the window coordinate system.

The current output position is set to be the point at the coordinates (x,y) . Nothing is drawn by the routine. The _lineto routine uses the current output position as the starting point when a line is drawn.

Note that the output position for graphics output differs from that for text output. The output position for text output can be set by use of the _settextposition routine.

Returns: The _moveto routines return the previous value of the output position for graphics.

See Also: _getcurrentposition, _lineto, _settextposition

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _moveto( 100, 100 )
call _lineto( 540, 100 )
call _lineto( 320, 380 )
call _lineto( 100, 100 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _moveto - DOS
 _moveto_w - DOS

Synopsis: subroutine _outgtext(text)
 character*(*) text

Description: The _outgtext routine displays the character string indicated by the argument *text*. The string must be terminated by a null character (char(0)).

The string is displayed starting at the current position (see the _moveto routine) in the current color and in the currently selected font (see the _setfont routine). The current position is updated to follow the displayed text.

When no font has been previously selected with _setfont, a default font will be used. The default font is an 8-by-8 bit-mapped font.

The graphics library can display text in three different ways.

1. The _outtext and _outmem routines can be used in any video mode. However, this variety of text can be displayed in only one size.
2. The _grtext routine displays text as a sequence of line segments, and can be drawn in different sizes, with different orientations and alignments.
3. The _outgtext routine displays text in the currently selected font. Both bit-mapped and vector fonts are supported; the size and type of text depends on the fonts that are available.

See Also: _registerfonts, _unregisterfonts, _setfont, _getfontinfo, _getgtextextent, _setgtextvector, _getgtextvector, _outtext, _outmem, _grtext

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i, n
character*10 buff

call _setvideomode( _VRES16COLOR )
n = _registerfonts( '*.fon'c )
do i = 0, n - 1
    write( buff, '( "n", i2.2, a1 )' ) i, char(0)
    call _setfont( buff )
    call _moveto( 100, 100 )
    call _outgtext( 'WATCOM Graphics'c )
    pause
    call _clearscreen( _GCLEARSCREEN )
enddo
call _unregisterfonts()
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _outmem(text, length)
 character*(*) text
 integer*2 length

Description: The _outmem routine displays the character string indicated by the argument *text*. The argument *length* specifies the number of characters to be displayed. Unlike the _outtext routine, _outmem will display the graphical representation of characters such as ASCII 10 and 0, instead of interpreting them as control characters.

The text is displayed using the current text color (see the _settextcolor routine), starting at the current text position (see the _settextposition routine). The text position is updated to follow the end of the displayed text.

The graphics library can display text in three different ways.

1. The _outtext and _outmem routines can be used in any video mode. However, this variety of text can be displayed in only one size.
2. The _grtext routine displays text as a sequence of line segments, and can be drawn in different sizes, with different orientations and alignments.
3. The _outgtext routine displays text in the currently selected font. Both bit-mapped and vector fonts are supported; the size and type of text depends on the fonts that are available.

See Also: _settextcolor, _settextposition, _settextwindow, _grtext, _outtext, _outgtext

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i
character*20 buf

call _clearscreen( _GCLEARSCREEN )
do i = 0, 255
    call _settextposition( 1 + mod( i, 16 ),
+                          1 + 5 * ( i / 16 ) )
    buf( 1:1 ) = char( i )
    call _outmem( buf, 1 )
enddo
pause
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _outtext(text)
 character*(*) text

Description: The _outtext routine displays the character string indicated by the argument *text*. The string must be terminated by a null character (char(0)). When a line-feed character (char(10)) is encountered in the string, the characters following will be displayed on the next row of the screen.

The text is displayed using the current text color (see the _settextcolor routine), starting at the current text position (see the _settextposition routine). The text position is updated to follow the end of the displayed text.

The graphics library can display text in three different ways.

1. The _outtext and _outmem routines can be used in any video mode. However, this variety of text can be displayed in only one size.
2. The _grtext routine displays text as a sequence of line segments, and can be drawn in different sizes, with different orientations and alignments.
3. The _outgtext routine displays text in the currently selected font. Both bit-mapped and vector fonts are supported; the size and type of text depends on the fonts that are available.

See Also: _settextcolor, _settextposition, _settextwindow, _grtext, _outmem,
 _outgtext

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _TEXTC80 )
call _settextposition( 10, 30 )
call _outtext( 'WATCOM Graphics'c )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis:

```
integer*2 function _pg_analyzechart( env, cat, values, n )
record /chartenv/ env
integer*4 cat(*)
real values(*)
integer*2 n

integer*2 function _pg_analyzechartms( env, cat, values,
                                     nseries, n,
                                     dim, labels )

record /chartenv/ env
integer*4 cat(*)
real values(*)
integer*2 nseries, n, dim
integer*4 labels(*)
```

Description: The `_pg_analyzechart` routines analyze either a single-series or a multi-series bar, column or line chart. These routines calculate default values for chart elements without actually displaying the chart.

The `_pg_analyzechart` routine analyzes a single-series bar, column or line chart. The chart environment structure `env` is filled with default values based on the type of chart and the values of the `cat` and `values` arguments. The arguments are the same as for the `_pg_chart` routine.

The `_pg_analyzechartms` routine analyzes a multi-series bar, column or line chart. The chart environment structure `env` is filled with default values based on the type of chart and the values of the `cat`, `values` and `labels` arguments. The arguments are the same as for the `_pg_chartms` routine.

Returns: The `_pg_analyzechart` routines return zero if successful; otherwise, a non-zero value is returned.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chart`, `_pg_chartpie`, `_pg_chartscatter`, `_pg_analyzepie`, `_pg_analyzescatter`

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+           _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
call _pg_analyzechart( env, categories,
+           values, NUM_VALUES )
! use manual scaling
env.yaxis.autoscale = 0
env.yaxis.scalemin = 0.0
env.yaxis.scalemax = 100.0
env.yaxis.ticinterval = 25.0
call _pg_chart( env, categories,
+           values, NUM_VALUES )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _pg_analyzechart - DOS
 _pg_analyzechartms - DOS

Synopsis: integer*2 function _pg_analyzepie(env, cat, values,
 explode, n)
 record /chartenv/ env
 integer*4 cat(*)
 real values(*)
 integer*2 explode(*), n

Description: The _pg_analyzepie routine analyzes a pie chart. This routine calculates default values for chart elements without actually displaying the chart.

 The chart environment structure *env* is filled with default values based on the values of the *cat*, *values* and *explode* arguments. The arguments are the same as for the _pg_chartpie routine.

Returns: The _pg_analyzepie routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_analyzechart, _pg_analyzescatter

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /
integer*2 explode( NUM_VALUES )
+           / 1, 0, 0, 0 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+           _PG_PIECHART, _PG_NOPERCENT )
env.maintitle.title = 'Pie Chart'c
env.legend.place = _PG_BOTTOM
call _pg_analyzepie( env, categories,
+           values, explode, NUM_VALUES )
! make legend window same width as data window
env.legend.autosize = 0
env.legend.legendwindow.x1 = env.datawindow.x1
env.legend.legendwindow.x2 = env.datawindow.x2
call _pg_chartpie( env, categories,
+           values, explode, NUM_VALUES )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics**Systems:** DOS

Synopsis:

```
integer*2 function _pg_analyzescatter( env, x, y, n )
record /chartenv/ env
real x(*), y(*)
integer*2 n

integer*2 function _pg_analyzescatterterms( env, x, y,
                                             nseries, n,
                                             dim, labels )

record /chartenv/ env
real x(*), y(*)
integer*2 nseries, n, dim
integer*4 labels(*)
```

Description: The `_pg_analyzescatter` routines analyze either a single-series or a multi-series scatter chart. These routines calculate default values for chart elements without actually displaying the chart.

The `_pg_analyzescatter` routine analyzes a single-series scatter chart. The chart environment structure `env` is filled with default values based on the values of the `x` and `y` arguments. The arguments are the same as for the `_pg_chartscatter` routine.

The `_pg_analyzescatterterms` routine analyzes a multi-series scatter chart. The chart environment structure `env` is filled with default values based on the values of the `x`, `y` and `labels` arguments. The arguments are the same as for the `_pg_chartscatterterms` routine.

Returns: The `_pg_analyzescatter` routines return zero if successful; otherwise, a non-zero value is returned.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chart`, `_pg_chartpie`,
`_pg_chartscatter`, `_pg_analyzechart`, `_pg_analyzepie`

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)
integer NUM_SERIES
parameter (NUM_SERIES=2)

integer*4 labels( NUM_SERIES )
real x( NUM_SERIES, NUM_VALUES )
+      / 5, 15, 30, 40, 10, 20, 30, 45 /
real y( NUM_SERIES, NUM_VALUES )
+      / 10, 15, 30, 45, 40, 30, 15, 5 /

record /chartenv/ env

labels( 1 ) = loc( 'Jan'c )
labels( 2 ) = loc( 'Feb'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+      _PG_SCATTERCHART, _PG_POINTANDLINE )
env.maintitle.title = 'Scatter Chart'c
call _pg_analyzescatterterms( env, x, y, NUM_SERIES,
+      NUM_VALUES, NUM_VALUES, labels )
! display x-axis labels with 2 decimal places
env.xaxis.autoscale = 0
env.xaxis.ticdecimals = 2
call _pg_chartscatterterms( env, x, y, NUM_SERIES,
+      NUM_VALUES, NUM_VALUES, labels )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _pg_analyzescatter - DOS
 _pg_analyzescatterterms - DOS

Synopsis:

```
integer*2 function _pg_chart( env, cat, values, n )
record /chartenv/ env
integer*4 cat(*)
real values(*)
integer*2 n

integer*2 function _pg_chartms( env, cat, values, nseries,
                                n, dim, labels )

record /chartenv/ env
integer*4 cat(*)
real values(*)
integer*2 nseries, n, dim
integer*4 labels(*)
```

Description: The `_pg_chart` routines display either a single-series or a multi-series bar, column or line chart. The type of chart displayed and other chart options are contained in the *env* argument. The argument *cat* is an array of addresses of strings. These strings describe the categories against which the data in the *values* array is charted.

The `_pg_chart` routine displays a bar, column or line chart from the single series of data contained in the *values* array. The argument *n* specifies the number of values to chart.

The `_pg_chartms` routine displays a multi-series bar, column or line chart. The argument *nseries* specifies the number of series of data to chart. The argument *values* is assumed to be a two-dimensional array defined as follows:

```
real values( nseries, dim )
```

The number of values used from each series is given by the argument *n*, where *n* is less than or equal to *dim*. The argument *labels* is an array of addresses of strings. These strings describe each of the series and are used in the chart legend.

Returns: The `_pg_chart` routines return zero if successful; otherwise, a non-zero value is returned.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chartpie`, `_pg_chartscatter`,
`_pg_analyzechart`, `_pg_analyzepie`, `_pg_analyzescatter`

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

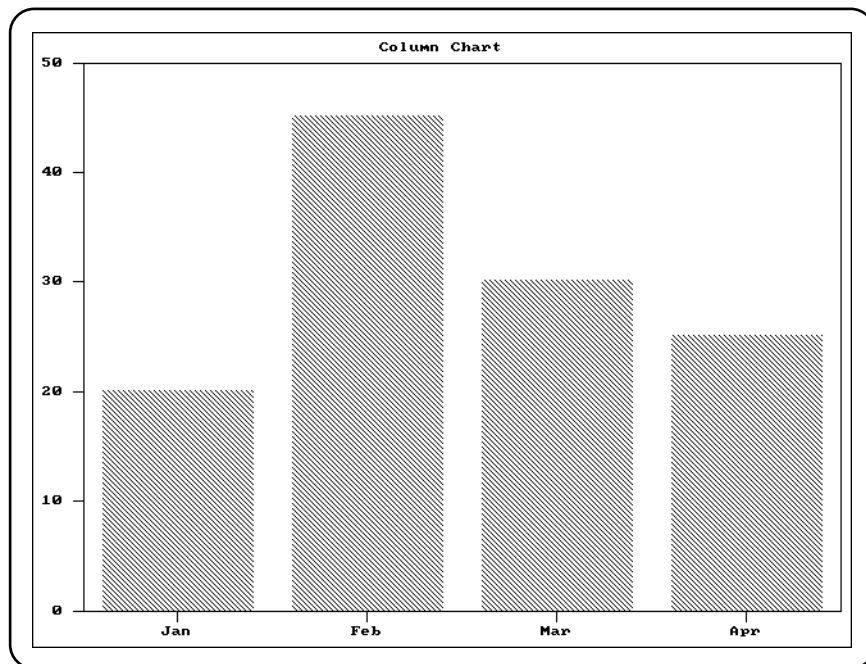
integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+                       _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
call _pg_chart( env, categories,
+               values, NUM_VALUES )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: _pg_chart - DOS
 _pg_chartms - DOS

Synopsis: integer*2 function _pg_chartpie(env, cat, values, explode, n)
 record /chartenv/ env
 integer*4 cat(*)
 real values(*)
 integer*2 explode(*), n

Description: The _pg_chartpie routine displays a pie chart. The chart is displayed using the options specified in the *env* argument.

The pie chart is created from the data contained in the *values* array. The argument *n* specifies the number of values to chart.

The argument *cat* is an array of addresses of strings. These strings describe each of the pie slices and are used in the chart legend. The argument *explode* is an array of values corresponding to each of the pie slices. For each non-zero element in the array, the corresponding pie slice is drawn "exploded", or slightly offset from the rest of the pie.

Returns: The _pg_chartpie routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartscatter,
 _pg_analyzechart, _pg_analyzepie, _pg_analyzescatter

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

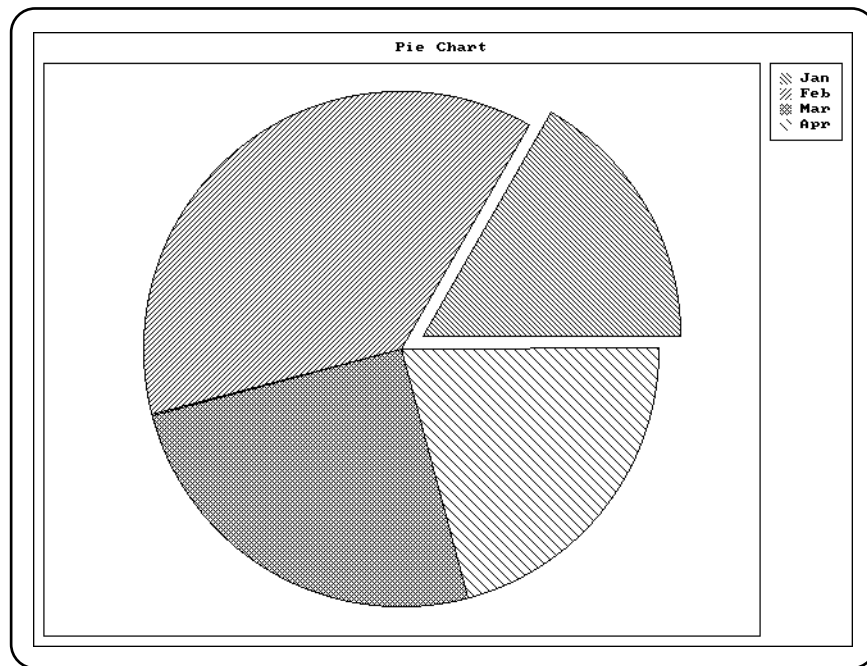
integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /
integer*2 explode( NUM_VALUES )
+           / 1, 0, 0, 0 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+                       _PG_PIECHART, _PG_NOPERCENT )
env.maintitle.title = 'Pie Chart'c
call _pg_chartpie( env, categories,
+                 values, explode, NUM_VALUES )
+
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

Synopsis:

```
integer*2 function _pg_chartscatter( env, x, y, n )
record /chartenv/ env
float x(*), y(*)
integer*2 n

integer*2 function _pg_chartscatterterms( env, x, y, nseries,
                                          n, dim, labels )

record /chartenv/ env
real x(*), y(*)
integer*2 nseries, n, dim
integer*4 labels(*)
```

Description: The `_pg_chartscatter` routines display either a single-series or a multi-series scatter chart. The chart is displayed using the options specified in the *env* argument.

The `_pg_chartscatter` routine displays a scatter chart from the single series of data contained in the arrays *x* and *y*. The argument *n* specifies the number of values to chart.

The `_pg_chartscatterterms` routine displays a multi-series scatter chart. The argument *nseries* specifies the number of series of data to chart. The arguments *x* and *y* are assumed to be two-dimensional arrays defined as follows:

```
real x( nseries, dim )
```

The number of values used from each series is given by the argument *n*, where *n* is less than or equal to *dim*. The argument *labels* is an array of addresses of strings. These strings describe each of the series and are used in the chart legend.

Returns: The `_pg_chartscatter` routines return zero if successful; otherwise, a non-zero value is returned.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chart`, `_pg_chartpie`,
`_pg_analyzechart`, `_pg_analyzepie`, `_pg_analyzescatter`

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)
integer NUM_SERIES
parameter (NUM_SERIES=2)

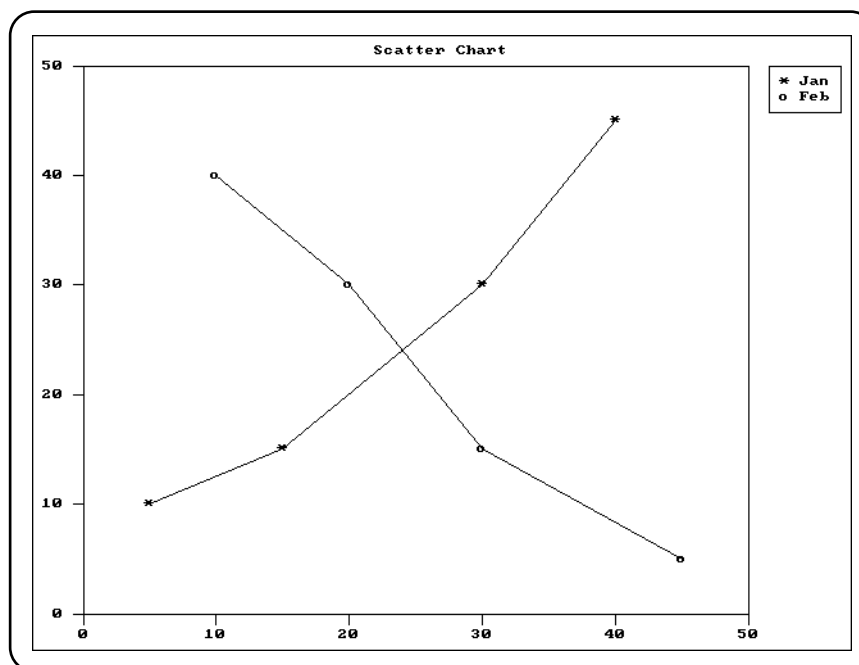
integer*4 labels( NUM_SERIES )
real x( NUM_SERIES, NUM_VALUES )
+      / 5, 15, 30, 40, 10, 20, 30, 45 /
real y( NUM_SERIES, NUM_VALUES )
+      / 10, 15, 30, 45, 40, 30, 15, 5 /

record /chartenv/ env

labels( 1 ) = loc( 'Jan'c )
labels( 2 ) = loc( 'Feb'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+      _PG_SCATTERCHART, _PG_POINTANDLINE )
env.maintitle.title = 'Scatter Chart'c
call _pg_chartscatter( env, x, y, NUM_SERIES,
+      NUM_VALUES, NUM_VALUES, labels )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



_pg_chartscatter Routines

Classification: PC Graphics

Systems: _pg_chartscatter - DOS
 _pg_chartscatterms - DOS

Synopsis: integer*2 function _pg_defaultchart(env, type, style)
 record /chartenv/ env
 integer*2 type, style

Description: The `_pg_defaultchart` routine initializes the chart structure *env* to contain default values before a chart is drawn. All values in the chart structure are initialized, including blanking of all titles. The chart type in the structure is initialized to the value *type*, and the chart style is initialized to *style*.

The argument *type* can have one of the following values:

<i>_PG_BARCHART</i>	Bar chart (horizontal bars)
<i>_PG_COLUMNCHART</i>	Column chart (vertical bars)
<i>_PG_LINECHART</i>	Line chart
<i>_PG_SCATTERCHART</i>	Scatter chart
<i>_PG_PIECHART</i>	Pie chart

Each type of chart can be drawn in one of two styles. For each chart type the argument *style* can have one of the following values:

Type	Style 1	Style 2
Bar	<code>_PG_PLAINBARS</code>	<code>_PG_STACKEDBARS</code>
Column	<code>_PG_PLAINBARS</code>	<code>_PG_STACKEDBARS</code>
Line	<code>_PG_POINTANDLINE</code>	<code>_PG_POINTONLY</code>
Scatter	<code>_PG_POINTANDLINE</code>	<code>_PG_POINTONLY</code>
Pie	<code>_PG_PERCENT</code>	<code>_PG_NOPERCENT</code>

For single-series bar and column charts, the chart style is ignored. The "plain" (clustered) and "stacked" styles only apply when there is more than one series of data. The "percent" style for pie charts causes percentages to be displayed beside each of the pie slices.

Returns: The `_pg_defaultchart` routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: `_pg_initchart`, `_pg_chart`, `_pg_chartpie`, `_pg_chartscatter`

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+           _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
call _pg_chart( env, categories,
+           values, NUM_VALUES )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _pg_getchardef(ch, def)
 integer*2 ch
 integer*1 def(8)

Description: The _pg_getchardef routine retrieves the current bit-map definition for the character *ch*. The bit-map is placed in the array *def*. The current font must be an 8-by-8 bit-mapped font.

Returns: The _pg_getchardef routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_setchardef

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

real x( NUM_VALUES )
+      / 5, 25, 45, 65 /
real y( NUM_VALUES )
+      / 5, 45, 25, 65 /
integer*1 diamond( 8 )
+      / '10'x, '28'x, '44'x, '82'x,
+      '44'x, '28'x, '10'x, '00'x /

record /chartenv/ env
integer*1 old_def( 8 )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+      _PG_SCATTERCHART, _PG_POINTANDLINE )
env.maintitle.title = 'Scatter Chart'c
! change asterisk character to diamond
call _pg_getchardef( ichar( '*' ), old_def )
call _pg_setchardef( ichar( '*' ), diamond )
call _pg_chartscatter( env, x, y, NUM_VALUES )
call _pg_setchardef( ichar( '*' ), old_def )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_pg_getchardef

Synopsis: integer*2 function _pg_getpalette(pal)
 record /paletteentry/ pal(*)

Description: The _pg_getpalette routine retrieves the internal palette of the presentation graphics system. The palette controls the colors, line styles, fill patterns and plot characters used to display each series of data in a chart.

The argument *pal* is an array of palette structures that will contain the palette. Each element of the palette is a structure containing the following fields:

<i>color</i>	color used to display series
<i>style</i>	line style used for line and scatter charts
<i>fill</i>	fill pattern used to fill interior of bar and pie sections
<i>plotchar</i>	character plotted on line and scatter charts

Returns: The _pg_getpalette routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_setpalette, _pg_resetpalette

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+       / 20, 45, 30, 25 /
integer*1 bricks( 8 )
+       / 'ff'x, '80'x, '80'x, '80'x,
+       'ff'x, '08'x, '08'x, '08'x /

record /chartenv/ env
record /paletteentry/ pal( _PG_PALETTELEN )
integer i

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+       _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! get default palette and change 1st entry
call _pg_getpalette( pal )
pal( 2 ).color = 12
do i = 1, 8
    pal( 2 ).fill( i ) = bricks( i )
enddo
! use new palette
call _pg_setpalette( pal )
call _pg_chart( env, categories,
+       values, NUM_VALUES )
! reset palette to default
call _pg_resetpalette()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _pg_getstyleset(style)
 integer*2 style(*)

Description: The _pg_getstyleset routine retrieves the internal style-set of the presentation graphics system. The style-set is a set of line styles used for drawing window borders and grid-lines. The argument *style* is an array that will contain the style-set.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_setstyleset, _pg_resetstyleset

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env
integer*2 style( _PG_PALETTELEN )

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+                     _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! turn on yaxis grid, and use style 2
env.yaxis.grid = 1
env.yaxis.gridstyle = 2
! get default style-set and change entry 2
call _pg_getstyleset( style )
style( 3 ) = '8888'x
! use new style-set
call _pg_setstyleset( style )
call _pg_chart( env, categories,
+             values, NUM_VALUES )
! reset style-set to default
call _pg_resetstyleset()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Description: The `_pg_hlabelchart` routine displays the text string *label* on the chart described by the *env* chart structure. The string is displayed horizontally starting at the point (x, y) , relative to the upper left corner of the chart. The *color* specifies the palette color used to display the string.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chart`, `_pg_chartpie`, `_pg_chartscluster`, `_pg_vlabelchart`

Classification: PC Graphics

90 Graphics Library Routines

Synopsis: integer*2 function _pg_initchart()

Description: The _pg_initchart routine initializes the presentation graphics system. This includes initializing the internal palette and style-set used when drawing charts. This routine must be called before any of the other presentation graphics routines.

The initialization of the presentation graphics system requires that a valid graphics mode has been selected. For this reason the _setvideomode routine must be called before _pg_initchart is called. If a font has been selected (with the _setfont routine), that font will be used when text is displayed in a chart. Font selection should also be done before initializing the presentation graphics system.

Returns: The _pg_initchart routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_chart, _pg_chartpie, _pg_chartscatter, _setvideomode, _setfont, _registerfonts

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+                       _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
call _pg_chart( env, categories,
+              values, NUM_VALUES )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_pg_resetpalette

Synopsis: integer*2 function _pg_resetpalette()

Description: The _pg_resetpalette routine resets the internal palette of the presentation graphics system to default values. The palette controls the colors, line styles, fill patterns and plot characters used to display each series of data in a chart. The default palette chosen is dependent on the current video mode.

Returns: The _pg_resetpalette routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_getpalette, _pg_setpalette

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+       / 20, 45, 30, 25 /
integer*1 bricks( 8 )
+       / 'ff'x, '80'x, '80'x, '80'x,
+       'ff'x, '08'x, '08'x, '08'x /

record /chartenv/ env
record /paletteentry/ pal( _PG_PALETTELEN )
integer i

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+       _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! get default palette and change 1st entry
call _pg_getpalette( pal )
pal( 2 ).color = 12
do i = 1, 8
    pal( 2 ).fill( i ) = bricks( i )
enddo
! use new palette
call _pg_setpalette( pal )
call _pg_chart( env, categories,
+       values, NUM_VALUES )
! reset palette to default
call _pg_resetpalette()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_pg_resetpalette

Synopsis: subroutine _pg_resetstyleset()

Description: The _pg_resetstyleset routine resets the internal style-set of the presentation graphics system to default values. The style-set is a set of line styles used for drawing window borders and grid-lines.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_getstyleset, _pg_setstyleset

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env
integer*2 style( _PG_PALETTELEN )

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+           _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! turn on yaxis grid, and use style 2
env.yaxis.grid = 1
env.yaxis.gridstyle = 2
! get default style-set and change entry 2
call _pg_getstyleset( style )
style( 3 ) = '8888'x
! use new style-set
call _pg_setstyleset( style )
call _pg_chart( env, categories,
+           values, NUM_VALUES )
! reset style-set to default
call _pg_resetstyleset()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _pg_setchardef(ch, def)
 integer*2 ch
 integer*1 def(8)

Description: The _pg_setchardef routine sets the current bit-map definition for the character *ch*. The bit-map is contained in the array *def*. The current font must be an 8-by-8 bit-mapped font.

Returns: The _pg_setchardef routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_getchardef

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

real x( NUM_VALUES )
+      / 5, 25, 45, 65 /
real y( NUM_VALUES )
+      / 5, 45, 25, 65 /
integer*1 diamond( 8 )
+      / '10'x, '28'x, '44'x, '82'x,
+      '44'x, '28'x, '10'x, '00'x /

record /chartenv/ env
integer*1 old_def( 8 )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+      _PG_SCATTERCHART, _PG_POINTANDLINE )
env.maintitle.title = 'Scatter Chart'c
! change asterisk character to diamond
call _pg_getchardef( ichar( '*' ), old_def )
call _pg_setchardef( ichar( '*' ), diamond )
call _pg_chartscatter( env, x, y, NUM_VALUES )
call _pg_setchardef( ichar( '*' ), old_def )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_pg_setchardef

Synopsis: integer*2 function _pg_setpalette(pal)
 record /paletteentry/ pal(*)

Description: The _pg_setpalette routine sets the internal palette of the presentation graphics system. The palette controls the colors, line styles, fill patterns and plot characters used to display each series of data in a chart.

The argument *pal* is an array of palette structures containing the new palette. Each element of the palette is a structure containing the following fields:

<i>color</i>	color used to display series
<i>style</i>	line style used for line and scatter charts
<i>fill</i>	fill pattern used to fill interior of bar and pie sections
<i>plotchar</i>	character plotted on line and scatter charts

Returns: The _pg_setpalette routine returns zero if successful; otherwise, a non-zero value is returned.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_getpalette, _pg_resetpalette

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+       / 20, 45, 30, 25 /
integer*1 bricks( 8 )
+       / 'ff'x, '80'x, '80'x, '80'x,
+       'ff'x, '08'x, '08'x, '08'x /

record /chartenv/ env
record /paletteentry/ pal( _PG_PALETTELEN )
integer i

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+       _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! get default palette and change 1st entry
call _pg_getpalette( pal )
pal( 2 ).color = 12
do i = 1, 8
    pal( 2 ).fill( i ) = bricks( i )
enddo
! use new palette
call _pg_setpalette( pal )
call _pg_chart( env, categories,
+       values, NUM_VALUES )
! reset palette to default
call _pg_resetpalette()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _pg_setstyleset(style)
 integer*2 style(*)

Description: The _pg_setstyleset routine retrieves the internal style-set of the presentation graphics system. The style-set is a set of line styles used for drawing window borders and grid-lines. The argument *style* is an array containing the new style-set.

See Also: _pg_defaultchart, _pg_initchart, _pg_chart, _pg_chartpie,
 _pg_chartscatter, _pg_getstyleset, _pg_resetstyleset

Example:

```
include 'graphapi.fi'
include 'graph.fi'
include 'pgapi.fi'
include 'pg.fi'

integer NUM_VALUES
parameter (NUM_VALUES=4)

integer*4 categories( NUM_VALUES )
real values( NUM_VALUES )
+           / 20, 45, 30, 25 /

record /chartenv/ env
integer*2 style( _PG_PALETTELEN )

categories( 1 ) = loc( 'Jan'c )
categories( 2 ) = loc( 'Feb'c )
categories( 3 ) = loc( 'Mar'c )
categories( 4 ) = loc( 'Apr'c )

call _setvideomode( _VRES16COLOR )
call _pg_initchart()
call _pg_defaultchart( env,
+                     _PG_COLUMNCHART, _PG_PLAINBARS )
env.maintitle.title = 'Column Chart'c
! turn on yaxis grid, and use style 2
env.yaxis.grid = 1
env.yaxis.gridstyle = 2
! get default style-set and change entry 2
call _pg_getstyleset( style )
style( 3 ) = '8888'x
! use new style-set
call _pg_setstyleset( style )
call _pg_chart( env, categories,
+              values, NUM_VALUES )
! reset style-set to default
call _pg_resetstyleset()
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Description: The `_pg_vlabelchart` routine displays the text string *label* on the chart described by the *env* chart structure. The string is displayed vertically starting at the point (x, y) , relative to the upper left corner of the chart. The *color* specifies the palette color used to display the string.

See Also: `_pg_defaultchart`, `_pg_initchart`, `_pg_chart`, `_pg_chartpie`, `_pg_chartscatter`, `_pg_hlabelchart`

Classification: PC Graphics

Synopsis:

```
integer*2 function _pie( fill, x1, y1, x2, y2,
                        x3, y3, x4, y4 )

integer*2 fill
integer*2 x1, y1
integer*2 x2, y2
integer*2 x3, y3
integer*2 x4, y4

integer*2 function _pie_w( fill, x1, y1, x2, y2,
                          x3, y3, x4, y4 )

integer*2 fill
double precision x1, y1
double precision x2, y2
double precision x3, y3
double precision x4, y4

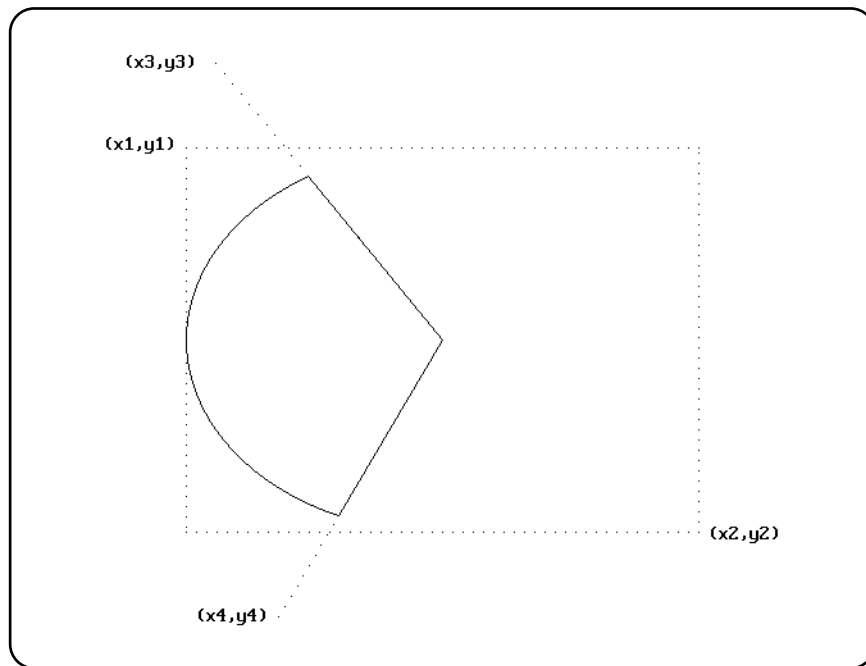
integer*2 function _pie_wxy( fill, p1, p2, p3, p4 )
integer*2 fill,
record /_wxycoord/ p1, p2
record /_wxycoord/ p3, p4
```

Description: The `_pie` routines draw pie-shaped wedges. The `_pie` routine uses the view coordinate system. The `_pie_w` and `_pie_wxy` routines use the window coordinate system.

The pie wedges are drawn by drawing an elliptical arc (in the way described for the `_arc` routines) and then joining the center of the rectangle that contains the ellipse to the two endpoints of the arc.

The elliptical arc is drawn with its center at the center of the rectangle established by the points $(x1, y1)$ and $(x2, y2)$. The arc is a segment of the ellipse drawn within this bounding rectangle. The arc starts at the point on this ellipse that intersects the vector from the centre of the ellipse to the point $(x3, y3)$. The arc ends at the point on this ellipse that intersects the vector from the centre of the ellipse to the point $(x4, y4)$. The arc is drawn in a counter-clockwise direction with the current plot action using the current color and the current line style.

The following picture illustrates the way in which the bounding rectangle and the vectors specifying the start and end points are defined.



When the coordinates $(x1, y1)$ and $(x2, y2)$ establish a line or a point (this happens when one or more of the x-coordinates or y-coordinates are equal), nothing is drawn.

The argument *fill* determines whether the figure is filled in or has only its outline drawn. The argument can have one of two values:

_GFILLINTERIOR fill the interior by writing pixels with the current plot action using the current color and the current fill mask

_GBORDER leave the interior unchanged; draw the outline of the figure with the current plot action using the current color and line style

Returns: The *_pie* routines return a non-zero value when the figure was successfully drawn; otherwise, zero is returned.

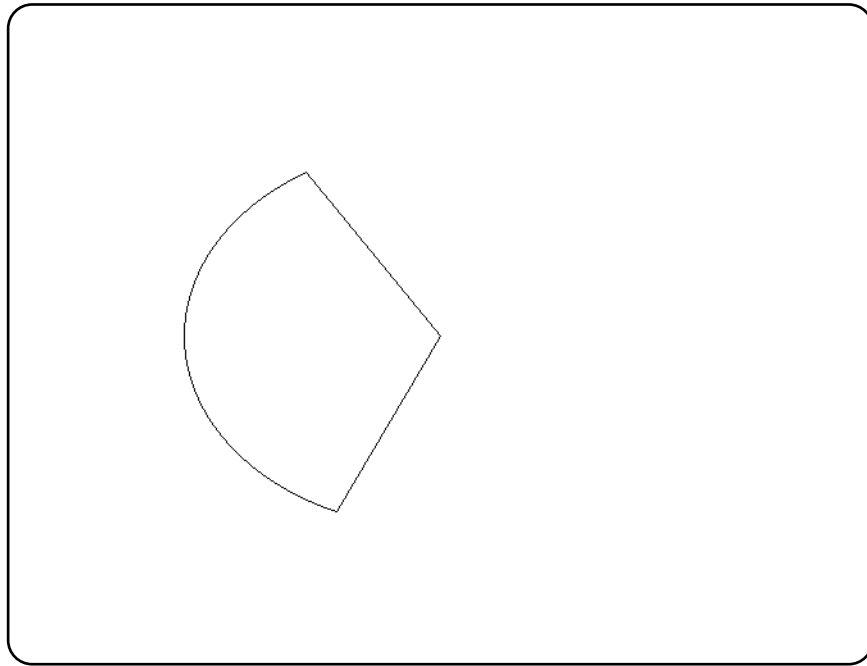
See Also: *_arc*, *_ellipse*, *_setcolor*, *_setfillmask*, *_setlinestyle*, *_setplotaction*

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _pie( _GBORDER, 120, 90, 520, 390,
+          140, 20, 190, 460 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: `_pie` - DOS
 `_pie_w` - DOS
 `_pie_wxy` - DOS

Synopsis:

```
integer*2 function _polygon( fill, numpts, points )
integer*2 fill
integer*2 numpts
record /xycoord/ points(*)

integer*2 function _polygon_w( fill, numpts, points )
integer*2 fill
integer*2 numpts
double precision points(*)

integer*2 function _polygon_wxy( fill, numpts, points )
integer*2 fill
integer*2 numpts
record /_wxycoord/ points(*)
```

Description: The `_polygon` routines draw polygons. The `_polygon` routine uses the view coordinate system. The `_polygon_w` and `_polygon_wxy` routines use the window coordinate system.

The polygon is defined as containing *numpts* points whose coordinates are given in the array *points*.

The argument *fill* determines whether the polygon is filled in or has only its outline drawn. The argument can have one of two values:

`_GFILLINTERIOR` fill the interior by writing pixels with the current plot action using the current color and the current fill mask

`_GBORDER` leave the interior unchanged; draw the outline of the figure with the current plot action using the current color and line style

Returns: The `_polygon` routines return a non-zero value when the polygon was successfully drawn; otherwise, zero is returned.

See Also: `_setcolor`, `_setfillmask`, `_setlinestyle`, `_setplotaction`

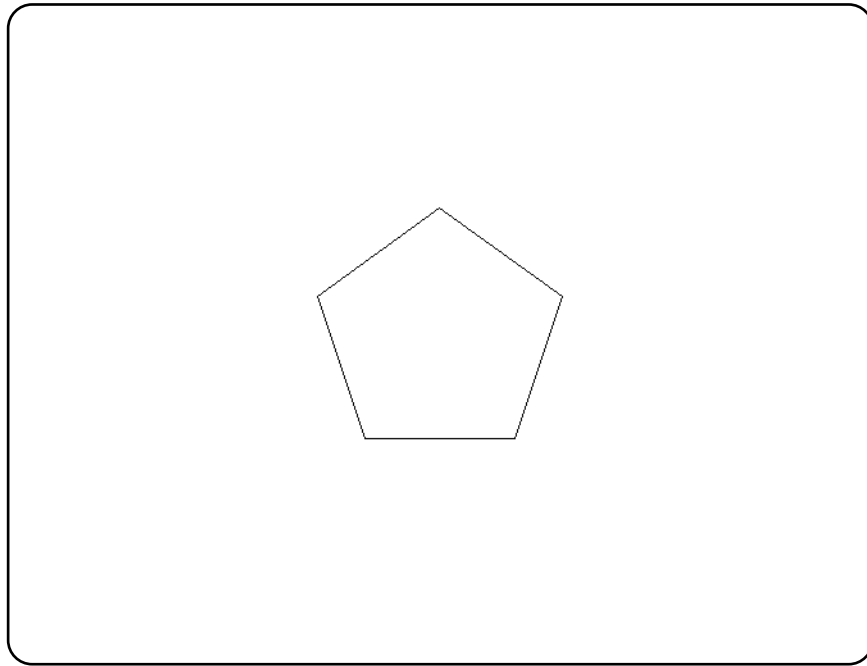
Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ points(5)/
+      319, 140, 224, 209, 261, 320,
+      378, 320, 415, 209/

call _setvideomode( _VRES16COLOR )
call _polygon( _GBORDER, 5, points )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: `_polygon` - DOS
 `_polygon_w` - DOS
 `_polygon_wxy` - DOS

Synopsis:

```
subroutine _putimage( x, y, image, mode )
integer*2 x, y
integer*1 image(*)
integer*2 mode

subroutine _putimage_w( x, y, image, mode )
double precision x, y
integer*1 image(*)
integer*2 mode
```

Description: The `_putimage` routines display the screen image indicated by the argument *image*. The `_putimage` routine uses the view coordinate system. The `_putimage_w` routine uses the window coordinate system.

The image is displayed upon the screen with its top left corner located at the point with coordinates (x, y) . The image was previously saved using the `_getimage` routines. The image is displayed in a rectangle whose size is the size of the rectangular image saved by the `_getimage` routines.

The image can be displayed in a number of ways, depending upon the value of the *mode* argument. This argument can have the following values:

<i>_GPSET</i>	replace the rectangle on the screen by the saved image
<i>_GPRESET</i>	replace the rectangle on the screen with the pixel values of the saved image inverted; this produces a negative image
<i>_GAND</i>	produce a new image on the screen by ANDing together the pixel values from the screen with those from the saved image
<i>_GOR</i>	produce a new image on the screen by ORing together the pixel values from the screen with those from the saved image
<i>_GXOR</i>	produce a new image on the screen by exclusive ORing together the pixel values from the screen with those from the saved image; the original screen is restored by two successive calls to the <code>_putimage</code> routine with this value, providing an efficient method to produce animated effects

See Also: `_getimage`, `_imagesize`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*1 image(:)
integer y, image_size, istat

call _setvideomode( _VRES16COLOR )
call _ellipse( _GFillInterior,
+             100, 100, 200, 200 )
image_size = _imagesize( 100, 100, 201, 201 )
allocate( image(image_size), stat = istat )
if( istat .eq. 0 )then
    call _getimage( 100, 100, 201, 201, image )
    call _putimage( 260, 200, image, _GPSET )
    call _putimage( 420, 100, image, _GPSET )
    do y = 100, 280, 20
        call _putimage( 420, y, image, _GXOR )
        call _putimage( 420, y+20, image, _GXOR )
    enddo
    deallocate( image )
endif
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _putimage - DOS
 _putimage_w - DOS

Synopsis:

```
integer*2 function _rectangle( fill, x1, y1, x2, y2 )
integer*2 fill
integer*2 x1, y1
integer*2 x2, y2

integer*2 function _rectangle_w( fill, x1, y1, x2, y2 )
integer*2 fill
double precision x1, y1
double precision x2, y2

integer*2 function _rectangle_wxy( fill, p1, p2 )
integer*2 fill
record /_wxycoord/ p1, p2
```

Description: The `_rectangle` routines draw rectangles. The `_rectangle` routine uses the view coordinate system. The `_rectangle_w` and `_rectangle_wxy` routines use the window coordinate system.

The rectangle is defined with opposite corners established by the points $(x1, y1)$ and $(x2, y2)$.

The argument *fill* determines whether the rectangle is filled in or has only its outline drawn. The argument can have one of two values:

`_GFILLINTERIOR` fill the interior by writing pixels with the current plot action using the current color and the current fill mask

`_GBORDER` leave the interior unchanged; draw the outline of the figure with the current plot action using the current color and line style

Returns: The `_rectangle` routines return a non-zero value when the rectangle was successfully drawn; otherwise, zero is returned.

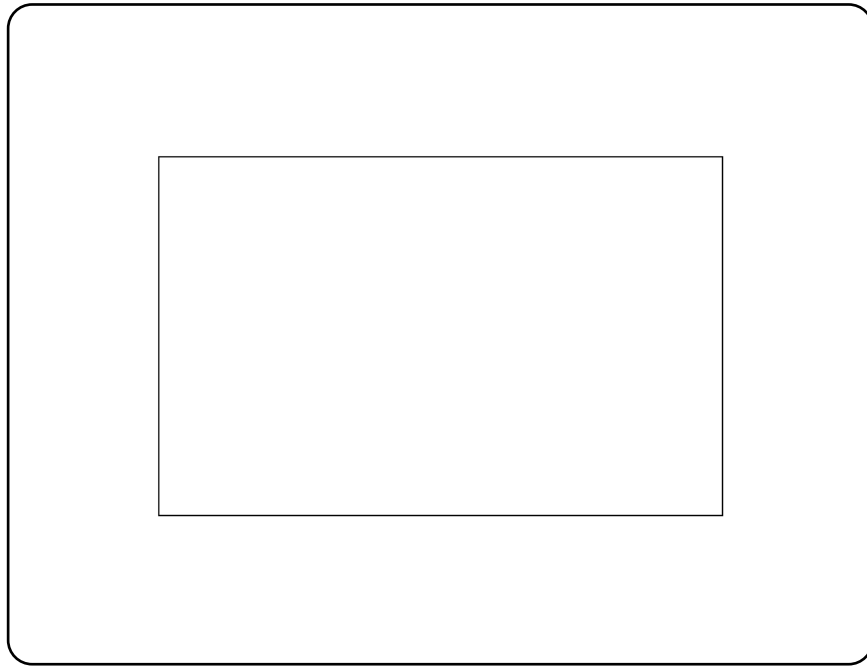
See Also: `_setcolor`, `_setfillmask`, `_setlinestyle`, `_setplotaction`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _rectangle( _GBORDER, 100, 100, 540, 380 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: `_rectangle` - DOS
 `_rectangle_w` - DOS
 `_rectangle_wxy` - DOS

Synopsis: integer*2 function _registerfonts(path)
 character*(*) path

Description: The _registerfonts routine initializes the font graphics system. Fonts must be registered, and a font selected, before text can be displayed with the _outgtext routine.

The argument *path* specifies the location of the font files. This argument is a file specification, and can contain drive and directory components and may contain wildcard characters. The _registerfonts routine opens each of the font files specified and reads the font information. Memory is allocated to store the characteristics of the font. These font characteristics are used by the _setfont routine when selecting a font.

Returns: The _registerfonts routine returns the number of fonts that were registered if the routine is successful; otherwise, a negative number is returned.

See Also: _unregisterfonts, _setfont, _getfontinfo, _outgtext, _getgtextextent,
 _setgtextvector, _getgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i, n
character*10 buff

call _setvideomode( _VRES16COLOR )
n = _registerfonts( '*.fon'c )
do i = 0, n - 1
  write( buff, '( "n", i2.2, a1 )' ) i, char(0)
  call _setfont( buff )
  call _moveto( 100, 100 )
  call _outgtext( 'WATCOM Graphics'c )
  pause
  call _clearscreen( _GCLEARSCREEN )
enddo
call _unregisterfonts()
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _remapallpalette(colors)
 integer*4 colors(*)

Description: The _remapallpalette routine sets (or remaps) all of the colors in the palette. The color values in the palette are replaced by the array of color values given by the argument *colors*. This routine is supported in all video modes, but only works with EGA, MCGA and VGA adapters.

The array *colors* must contain at least as many elements as there are supported colors. The newly mapped palette will cause the complete screen to change color wherever there is a pixel value of a changed color in the palette.

The representation of colors depends upon the hardware being used. The number of colors in the palette can be determined by using the _getvideoconfig routine.

Returns: The _remapallpalette routine returns (-1) if the palette is remapped successfully and zero otherwise.

See Also: _remappalette, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer colors(16)/
+   _BRIGHTWHITE, _YELLOW, _LIGHTMAGENTA,
+   _LIGHTRED, _LIGHTCYAN, _LIGHTGREEN,
+   _LIGHTBLUE, _GRAY, _WHITE, _BROWN,
+   _MAGENTA, _RED, _CYAN,
+   _GREEN, _BLUE, _BLACK/
integer x, y

call _setvideomode( _VRES16COLOR )
do y = 0, 3
  do x = 0, 3
    call _setcolor( x + 4 * y )
    call _rectangle( _GFILLINTERIOR,
+                   x * 160, y * 120,
+                   ( x + 1 ) * 160, ( y + 1 ) * 120 )
  enddo
enddo
pause
call _remapallpalette( colors )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*4 function _remappalette(pixval, color)
 integer*2 pixval
 integer*4 color

Description: The _remappalette routine sets (or remaps) the palette color *pixval* to be the color *color*. This routine is supported in all video modes, but only works with EGA, MCGA and VGA adapters.

The argument *pixval* is an index in the color palette of the current video mode. The argument *color* specifies the actual color displayed on the screen by pixels with pixel value *pixval*. Color values are selected by specifying the red, green and blue intensities that make up the color. Each intensity can be in the range from 0 to 63, resulting in 262144 possible different colors. A given color value can be conveniently specified as a value of type integer*4. The color value is of the form '00bbggrr'x, where bb is the blue intensity, gg is the green intensity and rr is the red intensity of the selected color. The file `graph.fi` defines constants containing the color intensities of each of the 16 default colors.

The _remappalette routine takes effect immediately. All pixels on the complete screen which have a pixel value equal to the value of *pixval* will now have the color indicated by the argument *color*.

Returns: The _remappalette routine returns the previous color for the pixel value if the palette is remapped successfully; otherwise, (-1) is returned.

See Also: _remapallpalette, _setvideomode

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer col
integer colors(16)/
+   _BLACK, _BLUE, _GREEN,
+   _CYAN, _RED, _MAGENTA,
+   _BROWN, _WHITE, _GRAY, _LIGHTBLUE,
+   _LIGHTGREEN, _LIGHTCYAN, _LIGHTRED,
+   _LIGHTMAGENTA, _YELLOW, _BRIGHTWHITE/

call _setvideomode( _VRES16COLOR )
do col = 1, 16
    call _remappalette( 0, colors(col) )
    pause
enddo
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_scrolltextwindow

Synopsis: subroutine _scrolltextwindow(rows)
 integer*2 rows

Description: The _scrolltextwindow routine scrolls the lines in the current text window. A text window is defined with the _settextwindow routine. By default, the text window is the entire screen.

The argument *rows* specifies the number of rows to scroll. A positive value means to scroll the text window up or towards the top of the screen. A negative value means to scroll the text window down or towards the bottom of the screen. Specifying a number of rows greater than the height of the text window is equivalent to clearing the text window with the _clearscreen routine.

Two constants are defined that can be used with the _scrolltextwindow routine:

_GSCROLLUP the contents of the text window are scrolled up (towards the top of the screen) by one row

_GSCROLLEDOWN the contents of the text window are scrolled down (towards the bottom of the screen) by one row

See Also: _settextwindow, _clearscreen, _outtext, _outmem, _settextposition

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i
character*80 buff

call _setvideomode( _TEXTC80 )
call _settextwindow( 5, 20, 20, 40 )
do i = 1, 10
    write( buff, '( "Line ", i2, a1, a1 )' )
+      i, char(10), char(0)
    call _outtext( buff )
enddo
pause
call _scrolltextwindow( _GSCROLLEDOWN )
pause
call _scrolltextwindow( _GSCROLLUP )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_selectpalette

Synopsis: integer*2 function _selectpalette(palnum)
 integer*2 palnum

Description: The _selectpalette routine selects the palette indicated by the argument *palnum* from the color palettes available. This routine is only supported by the video modes _MRES4COLOR and _MRESNOCOLOR.

Mode _MRES4COLOR supports four palettes of four colors. In each palette, color 0, the background color, can be any of the 16 possible colors. The color values associated with the other three pixel values, (1, 2 and 3), are determined by the selected palette.

The following table outlines the available color palettes:

Palette Number	1	Pixel Values 2	3
0	green	red	brown
1	cyan	magenta	white
2	light green	light red	yellow
3	light cyan	light magenta	bright white

Returns: The _selectpalette routine returns the number of the previously selected palette.

See Also: _setvideomode, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer x, y, pal

call _setvideomode( _MRES4COLOR )
do y = 0, 1
  do x = 0, 1
    call _setcolor( x + 2 * y )
    call _rectangle( _GFILLINTERIOR,
+                   x * 160, y * 100,
+                   ( x + 1 ) * 160, ( y + 1 ) * 100 )
  enddo
enddo
do pal = 0, 3
  call _selectpalette( pal )
  pause
enddo
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _setactivepage(pagenum)
 integer*2 pagenum

Description: The _setactivepage routine selects the page (in memory) to which graphics output is written. The page to be selected is given by the *pagenum* argument.

Only some combinations of video modes and hardware allow multiple pages of graphics to exist. When multiple pages are supported, the active page may differ from the visual page. The graphics information in the visual page determines what is displayed upon the screen. Animation may be accomplished by alternating the visual page. A graphics page can be constructed without affecting the screen by setting the active page to be different than the visual page.

The number of available video pages can be determined by using the _getvideoconfig routine. The default video page is 0.

Returns: The _setactivepage routine returns the number of the previous page when the active page is set successfully; otherwise, a negative number is returned.

See Also: _getactivepage, _setvisualpage, _getvisualpage, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_apage, old_vpage

call _setvideomode( _HRES16COLOR )
old_apage = _getactivepage()
old_vpage = _getvisualpage()
! draw an ellipse on page 0
call _setactivepage( 0 )
call _setvisualpage( 0 )
call _ellipse( _GFillInterior, 100, 50,
+              540, 150 )
! draw a rectangle on page 1
call _setactivepage( 1 )
call _rectangle( _GFillInterior, 100, 50,
+               540, 150 )
pause
! display page 1
call _setvisualpage( 1 )
pause
call _setactivepage( old_apage )
call _setvisualpage( old_vpage )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_setbkcolor

Synopsis: integer*4 function _setbkcolor(color)
 integer*4 color

Description: The _setbkcolor routine sets the current background color to be that of the *color* argument. In text modes, the background color controls the area behind each individual character. In graphics modes, the background refers to the entire screen. The default background color is 0.

When the current video mode is a graphics mode, any pixels with a zero pixel value will change to the color of the *color* argument. When the current video mode is a text mode, nothing will immediately change; only subsequent output is affected.

Returns: The _setbkcolor routine returns the previous background color.

See Also: _getbkcolor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer bk, old_bk
integer colors(16)/
+   _BLACK, _BLUE, _GREEN,
+   _CYAN, _RED, _MAGENTA,
+   _BROWN, _WHITE, _GRAY, _LIGHTBLUE,
+   _LIGHTGREEN, _LIGHTCYAN, _LIGHTRED,
+   _LIGHTMAGENTA, _YELLOW, _BRIGHTWHITE/

call _setvideomode( _VRES16COLOR )
old_bk = _getbkcolor()
do bk = 1, 16
    call _setbkcolor( colors( bk ) )
    pause
enddo
call _setbkcolor( old_bk )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _setcharsize(height, width)
 integer*2 height, width

 subroutine _setcharsize_w(height, width)
 double precision height, width

Description: The `_setcharsize` routines set the character height and width to the values specified by the arguments *height* and *width*. For the `_setcharsize` routine, the arguments *height* and *width* represent a number of pixels. For the `_setcharsize_w` routine, the arguments *height* and *width* represent lengths along the y-axis and x-axis in the window coordinate system.

These sizes are used when displaying text with the `_grtext` routine. The default character sizes are dependent on the graphics mode selected, and can be determined by the `_gettextsettings` routine.

See Also: `_grtext`, `_gettextsettings`

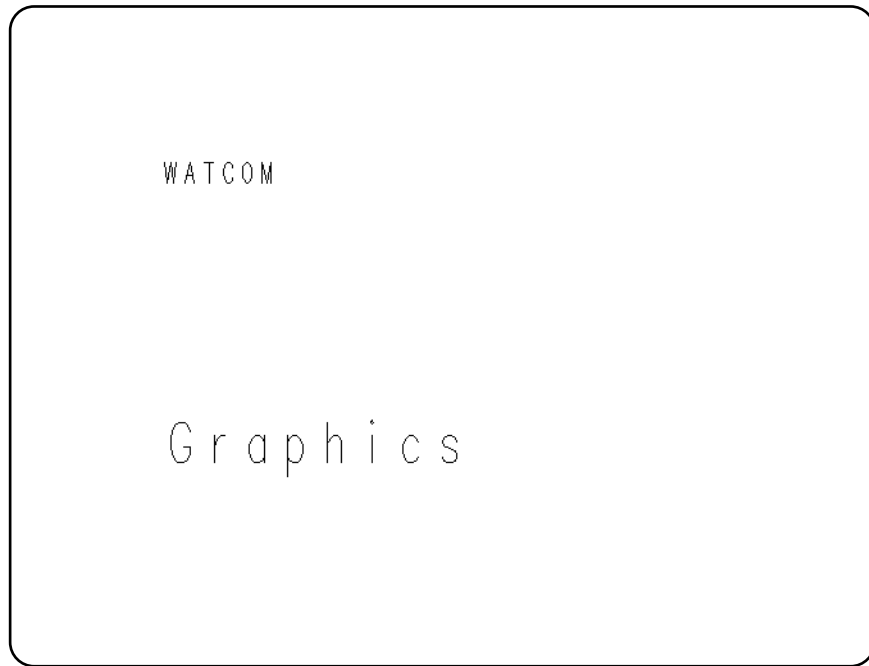
Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /textsettings/ ts

call _setvideomode( _VRES16COLOR )
call _gettextsettings( ts )
call _grtext( 100, 100, 'WATCOM'c )
call _setcharsize( 2 * ts.height, 2 * ts.width )
call _grtext( 100, 300, 'Graphics'c )
call _setcharsize( ts.height, ts.width )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: _setcharsize - DOS
 _setcharsize_w - DOS

Synopsis: subroutine _setcharspacing(space)
 integer*2 space

 subroutine _setcharspacing_w(space)
 double precision space

Description: The _setcharspacing routines set the current character spacing to have the value of the argument *space*. For the _setcharspacing routine, *space* represents a number of pixels. For the _setcharspacing_w routine, *space* represents a length along the x-axis in the window coordinate system.

The character spacing specifies the additional space to leave between characters when a text string is displayed with the _grtext routine. A negative value can be specified to cause the characters to be drawn closer together. The default value of the character spacing is 0.

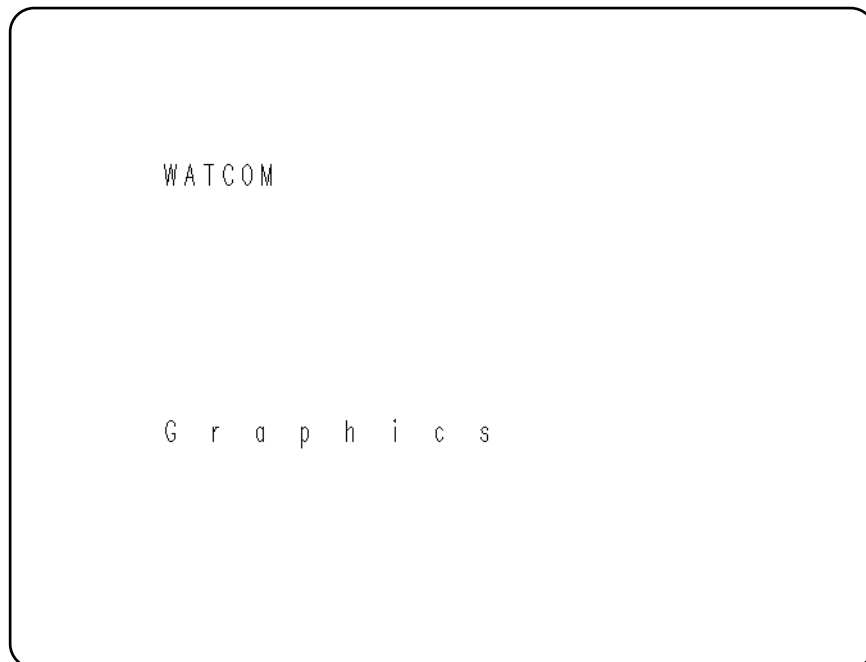
See Also: _grtext, _gettextsettings

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _grtext( 100, 100, 'WATCOM'c )
call _setcharspacing( 20 )
call _grtext( 100, 300, 'Graphics'c )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

_setcharspacing Routines

Systems: _setcharspacing - DOS
 _setcharspacing_w - DOS

Synopsis: subroutine _setcliprgn(x1, y1, x2, y2)
 integer*2 x1, y1
 integer*2 x2, y2

Description: The _setcliprgn routine restricts the display of graphics output to the clipping region. This region is a rectangle whose opposite corners are established by the physical points (x1,y1) and (x2,y2) .

The _setcliprgn routine does not affect text output using the _outtext and _outmem routines. To control the location of text output, see the _settextwindow routine.

See Also: _settextwindow, _setvieworg, _setviewport

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*2 x1, y1, x2, y2

call _setvideomode( _VRES16COLOR )
call _getcliprgn( x1, y1, x2, y2 )
call _setcliprgn( 130, 100, 510, 380 )
call _ellipse( _GBORDER, 120, 90, 520, 390 )
pause
call _setcliprgn( x1, y1, x2, y2 )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_setcolor

Synopsis: integer*2 function _setcolor(pixval)
 integer*2 pixval

Description: The `_setcolor` routine sets the pixel value for the current color to be that indicated by the *pixval* argument. The current color is only used by the routines that produce graphics output; text output with `_outtext` uses the current text color (see the `_settextcolor` routine). The default color value is one less than the maximum number of colors in the current video mode.

Returns: The `_setcolor` routine returns the previous value of the current color.

See Also: `_getcolor`, `_settextcolor`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer col, old_col

call _setvideomode( _VRES16COLOR )
old_col = _getcolor()
do col = 0, 15
    call _setcolor( col )
    call _rectangle( _GFILLINTERIOR,
+                  100, 100, 540, 380 )
    pause
enddo
call _setcolor( old_col )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _setfillmask(mask)
 integer*1 mask(8)

Description: The _setfillmask routine sets the current fill mask to the value of the argument *mask*.

The fill mask is an eight-byte array which is interpreted as a square pattern (8 by 8) of 64 bits. Each bit in the mask corresponds to a pixel. When a region is filled, each point in the region is mapped onto the fill mask. When a bit from the mask is one, the pixel value of the corresponding point is set using the current plotting action with the current color; when the bit is zero, the pixel value of that point is not affected.

When the fill mask is not set, a fill operation will set all points in the fill region to have a pixel value of the current color. By default, no fill mask is set.

See Also: _getfillmask, _ellipse, _floodfill, _rectangle, _polygon, _pie, _setcolor, _setplotaction

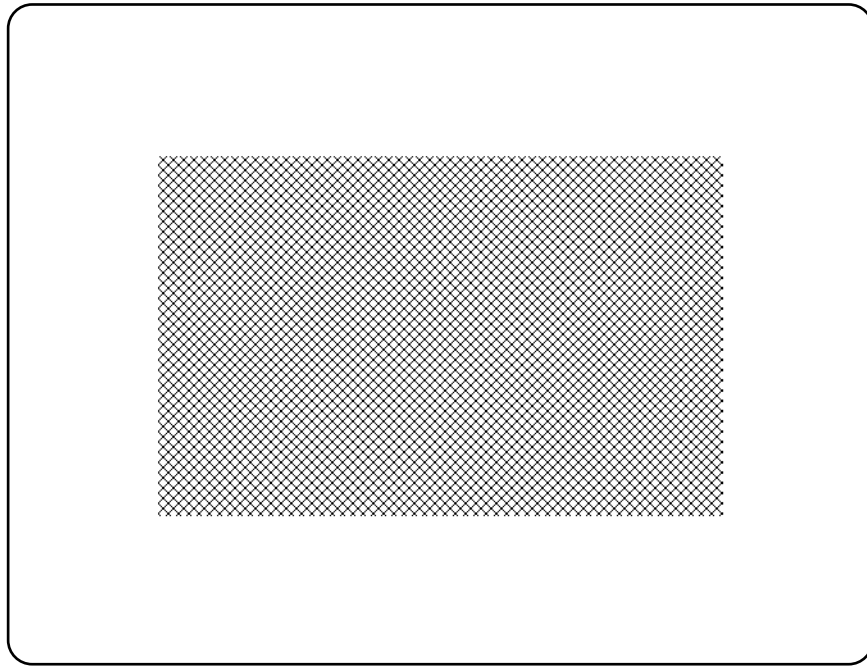
Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*1 old_mask(8)
integer*1 new_mask(8)/
+           '81'x, '42'x, '24'x, '18'x,
+           '18'x, '24'x, '42'x, '81'x/

call _setvideomode( _VRES16COLOR )
call _getfillmask( old_mask )
call _setfillmask( new_mask )
call _rectangle( _GFILLINTERIOR,
+               100, 100, 540, 380 )
call _setfillmask( old_mask )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _setfont(opt)
 character*(*) opt

Description: The _setfont routine selects a font from the list of registered fonts (see the _registerfonts routine). The font selected becomes the current font and is used whenever text is displayed with the _outgtext routine. The routine will fail if no fonts have been registered, or if a font cannot be found that matches the given characteristics.

The argument *opt* is a string of characters specifying the characteristics of the desired font. These characteristics determine which font is selected. The options may be separated by blanks and are not case-sensitive. Any number of options may be specified and in any order. The available options are:

<i>hX</i>	character height X (in pixels)
<i>wX</i>	character width X (in pixels)
<i>f</i>	choose a fixed-width font
<i>p</i>	choose a proportional-width font
<i>r</i>	choose a raster (bit-mapped) font
<i>v</i>	choose a vector font
<i>b</i>	choose the font that best matches the options
<i>nX</i>	choose font number X (the number of fonts is returned by the _registerfonts routine)
<i>t'facename'</i>	choose a font with specified facename

The facename option is specified as a "t" followed by a facename enclosed in single quotes. The available facenames are:

<i>Courier</i>	fixed-width raster font with serifs
<i>Helv</i>	proportional-width raster font without serifs
<i>Tms Rmn</i>	proportional-width raster font with serifs
<i>Script</i>	proportional-width vector font that appears similar to hand-writing
<i>Modern</i>	proportional-width vector font without serifs
<i>Roman</i>	proportional-width vector font with serifs

When "nX" is specified to select a particular font, the other options are ignored.

If the best fit option ("b") is specified, _setfont will always be able to select a font. The font chosen will be the one that best matches the options specified. The following precedence is given to the options when selecting a font:

1. Pixel height (higher precedence is given to heights less than the specified height)

2. Facename
3. Pixel width
4. Font type (fixed or proportional)

When a pixel height or width does not match exactly and a vector font has been selected, the font will be stretched appropriately to match the given size.

Returns: The `_setfont` routine returns zero if successful; otherwise, (-1) is returned.

See Also: `_registerfonts`, `_unregisterfonts`, `_getfontinfo`, `_outgtext`, `_getgtexttextent`, `_setgtextvector`, `_getgtextvector`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i, n
character*10 buff

call _setvideomode( _VRES16COLOR )
n = _registerfonts( '*.fon'c )
do i = 0, n - 1
    write( buff, '( "n", i2.2, a1 )' ) i, char(0)
    call _setfont( buff )
    call _moveto( 100, 100 )
    call _outgtext( 'WATCOM Graphics'c )
    pause
    call _clearscreen( _GCLEARSCREEN )
enddo
call _unregisterfonts()
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: record /xycoord/ function _setgtextvector(x, y)
 integer*2 x, y

Description: The _setgtextvector routine sets the orientation for text output used by the _outgtext routine to the vector specified by the arguments (x,y) . Each of the arguments can have a value of -1, 0 or 1, allowing for text to be displayed at any multiple of a 45-degree angle. The default text orientation, for normal left-to-right text, is the vector (1,0) .

Returns: The _setgtextvector routine returns, as an xycoord structure, the previous value of the text orientation vector.

See Also: _registerfonts, _unregisterfonts, _setfont, _getfontinfo, _outgtext,
 _getgtexttextent, _getgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /xycoord/ old_vec

call _setvideomode( _VRES16COLOR )
old_vec = _getgtextvector()
call _setgtextvector( 0, -1 )
call _moveto( 100, 100 )
call _outgtext( 'WATCOM Graphics'c )
call _setgtextvector( old_vec.xcoord, old_vec.ycoord )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _setlinestyle(style)
 integer*2 style

Description: The _setlinestyle routine sets the current line-style mask to the value of the *style* argument.

The line-style mask determines the style by which lines and arcs are drawn. The mask is treated as an array of 16 bits. As a line is drawn, a pixel at a time, the bits in this array are cyclically tested. When a bit in the array is 1, the pixel value for the current point is set using the current color according to the current plotting action; otherwise, the pixel value for the point is left unchanged. A solid line would result from a value of 'FFFF'x and a dashed line would result from a value of 'F0F0'x.

The default line style mask is 'FFFF'x.

See Also: _getlinestyle, _lineto, _rectangle, _polygon, _setplotaction

Example:

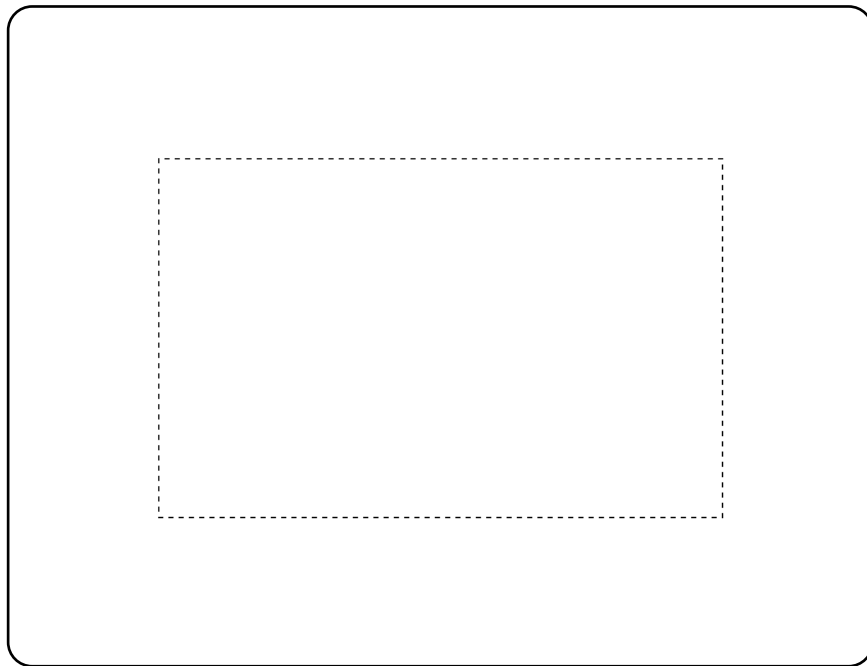
```
include 'graphapi.fi'
include 'graph.fi'

integer DASHED
parameter (DASHED='f0f0'x)

integer old_style

call _setvideomode( _VRES16COLOR )
old_style = _getlinestyle()
call _setlinestyle( DASHED )
call _rectangle( _GBORDER, 100, 100, 540, 380 )
call _setlinestyle( old_style )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

_setpixel Routines

Synopsis: integer*2 function _setpixel(x, y)
 integer*2 x, y

 integer*2 function _setpixel_w(x, y)
 double precision x, y

Description: The _setpixel routine sets the pixel value of the point (x,y) using the current plotting action with the current color. The _setpixel routine uses the view coordinate system. The _setpixel_w routine uses the window coordinate system.

A pixel value is associated with each point. The values range from 0 to the number of colors (less one) that can be represented in the palette for the current video mode. The color displayed at the point is the color in the palette corresponding to the pixel number. For example, a pixel value of 3 causes the fourth color in the palette to be displayed at the point in question.

Returns: The _setpixel routines return the previous value of the indicated pixel if the pixel value can be set; otherwise, (-1) is returned.

See Also: _getpixel, _setcolor, _setplotaction

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer x, y, i
real urand
integer seed

seed = 75347
call _setvideomode( _VRES16COLOR )
call _rectangle( _GBORDER, 100, 100, 540, 380 )
do i = 0, 60000
  x = 101 + mod( int( urand( seed )*32767 ),
+               439 )
  y = 101 + mod( int( urand( seed )*32767 ),
+               279 )
  call _setcolor( _getpixel( x, y ) + 1 )
  call _setpixel( x, y )
enddo
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: _setpixel - DOS
 _setpixel_w - DOS

_setplotaction

Synopsis: integer*2 function _setplotaction(action)
 integer*2 action

Description: The _setplotaction routine sets the current plotting action to the value of the *action* argument.

The drawing routines cause pixels to be set with a pixel value. By default, the value to be set is obtained by replacing the original pixel value with the supplied pixel value. Alternatively, the replaced value may be computed as a function of the original and the supplied pixel values.

The plotting action can have one of the following values:

<u>_GPSET</u>	replace the original screen pixel value with the supplied pixel value
<u>_GAND</u>	replace the original screen pixel value with the <i>bitwise and</i> of the original pixel value and the supplied pixel value
<u>_GOR</u>	replace the original screen pixel value with the <i>bitwise or</i> of the original pixel value and the supplied pixel value
<u>_GXOR</u>	replace the original screen pixel value with the <i>bitwise exclusive-or</i> of the original pixel value and the supplied pixel value. Performing this operation twice will restore the original screen contents, providing an efficient method to produce animated effects.

Returns: The previous value of the plotting action is returned.

See Also: _getplotaction

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_act

call _setvideomode( _VRES16COLOR )
old_act = _getplotaction()
call _setplotaction( _GPSET )
call _rectangle( _GFILLINTERIOR, 100, 100,
+               540, 380 )
pause
call _setplotaction( _GXOR )
call _rectangle( _GFILLINTERIOR, 100, 100,
+               540, 380 )
pause
call _setplotaction( old_act )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _settextalign(horiz, vert)
 integer*2 horiz, vert

Description: The _settextalign routine sets the current text alignment to the values specified by the arguments *horiz* and *vert*. When text is displayed with the _grtext routine, it is aligned (justified) horizontally and vertically about the given point according to the current text alignment settings.

The horizontal component of the alignment can have one of the following values:

<u>NORMAL</u>	use the default horizontal alignment for the current setting of the text path
<u>LEFT</u>	the text string is left justified at the given point
<u>CENTER</u>	the text string is centred horizontally about the given point
<u>RIGHT</u>	the text string is right justified at the given point

The vertical component of the alignment can have one of the following values:

<u>NORMAL</u>	use the default vertical alignment for the current setting of the text path
<u>TOP</u>	the top of the text string is aligned at the given point
<u>CAP</u>	the cap line of the text string is aligned at the given point
<u>HALF</u>	the text string is centred vertically about the given point
<u>BASE</u>	the base line of the text string is aligned at the given point
<u>BOTTOM</u>	the bottom of the text string is aligned at the given point

The default is to use LEFT alignment for the horizontal component unless the text path is PATH_LEFT, in which case RIGHT alignment is used. The default value for the vertical component is TOP unless the text path is PATH_UP, in which case BOTTOM alignment is used.

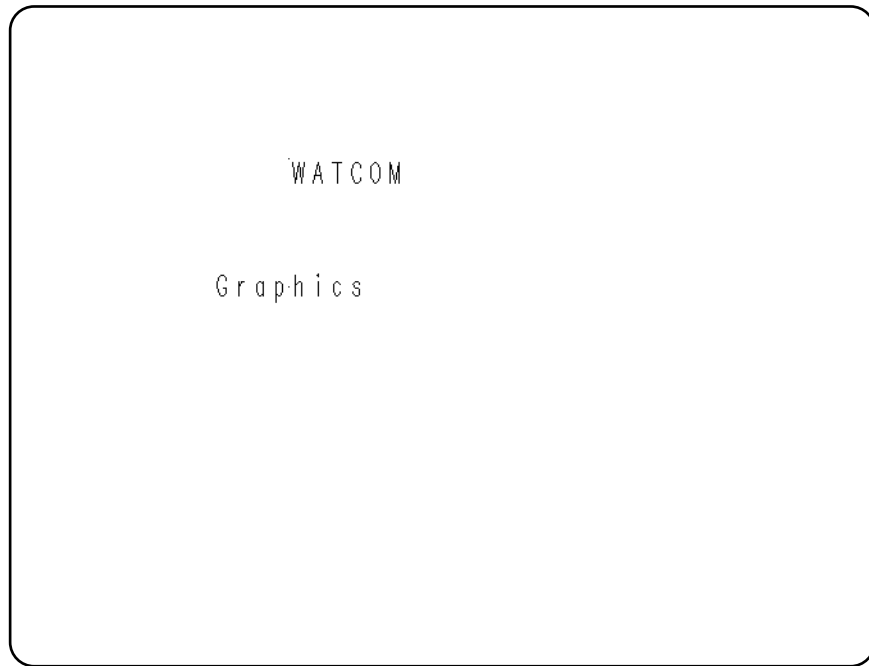
See Also: _grtext, _gettextsettings

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _grtext( 200, 100, 'WATCOM'c )
call _setpixel( 200, 100 )
call _settextalign( _CENTER, _HALF )
call _grtext( 200, 200, 'Graphics'c )
call _setpixel( 200, 200 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

_settextcolor

Synopsis: integer*2 function _settextcolor(pixval)
 integer*2 pixval

Description: The _settextcolor routine sets the current text color to be the color indicated by the pixel value of the *pixval* argument. This is the color value used for displaying text with the _outtext and _outmem routines. Use the _setcolor routine to change the color of graphics output. The default text color value is set to 7 whenever a new video mode is selected.

The pixel value *pixval* is a number in the range 0-31. Colors in the range 0-15 are displayed normally. In text modes, blinking colors are specified by adding 16 to the normal color values. The following table specifies the default colors in color text modes.

Pixel value	Color	Pixel value	Color
0	Black	8	Gray
1	Blue	9	Light Blue
2	Green	10	Light Green
3	Cyan	11	Light Cyan
4	Red	12	Light Red
5	Magenta	13	Light Magenta
6	Brown	14	Yellow
7	White	15	Bright White

Returns: The _settextcolor routine returns the pixel value of the previous text color.

See Also: _gettextcolor, _outtext, _outmem, _setcolor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_col
integer old_bk

call _setvideomode( _TEXTC80 )
old_col = _gettextcolor()
old_bk = _getbkcolor()
call _settextcolor( 7 )
call _setbkcolor( _BLUE )
call _outtext( ' WATCOM '//char(10)//
+             'Graphics'c )
call _settextcolor( old_col )
call _setbkcolor( old_bk )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_settextcursor

Synopsis: integer*2 function _settextcursor(cursor)
 integer*2 cursor

Description: The _settextcursor routine sets the attribute, or shape, of the cursor in text modes. The argument *cursor* specifies the new cursor shape. The cursor shape is selected by specifying the top and bottom rows in the character matrix. The high byte of *cursor* specifies the top row of the cursor; the low byte specifies the bottom row.

Some typical values for *cursor* are:

Cursor	Shape
'0607'x	normal underline cursor
'0007'x	full block cursor
'0407'x	half-height block cursor
'2000'x	no cursor

Returns: The _settextcursor routine returns the previous cursor shape when the shape is set successfully; otherwise, (-1) is returned.

See Also: _gettextcursor, _displaycursor

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer*2 old_shape

old_shape = _gettextcursor()
call _settextcursor( '0007'x )
call _outtext(
+   char(10)//'Block cursor'c )
pause
call _settextcursor( '0407'x )
call _outtext(
+   char(10)//'Half height cursor'c )
pause
call _settextcursor( '2000'x )
call _outtext(
+   char(10)//'No cursor'c )
pause
end
```

Classification: PC Graphics

Systems: DOS

_settextorient

Synopsis: subroutine _settextorient(vecx, vecy)
 integer*2 vecx, vecy

Description: The _settextorient routine sets the current text orientation to the vector specified by the arguments (vecx,vecy) . The text orientation specifies the direction of the base-line vector when a text string is displayed with the _grtext routine. The default text orientation, for normal left-to-right text, is the vector (1,0) .

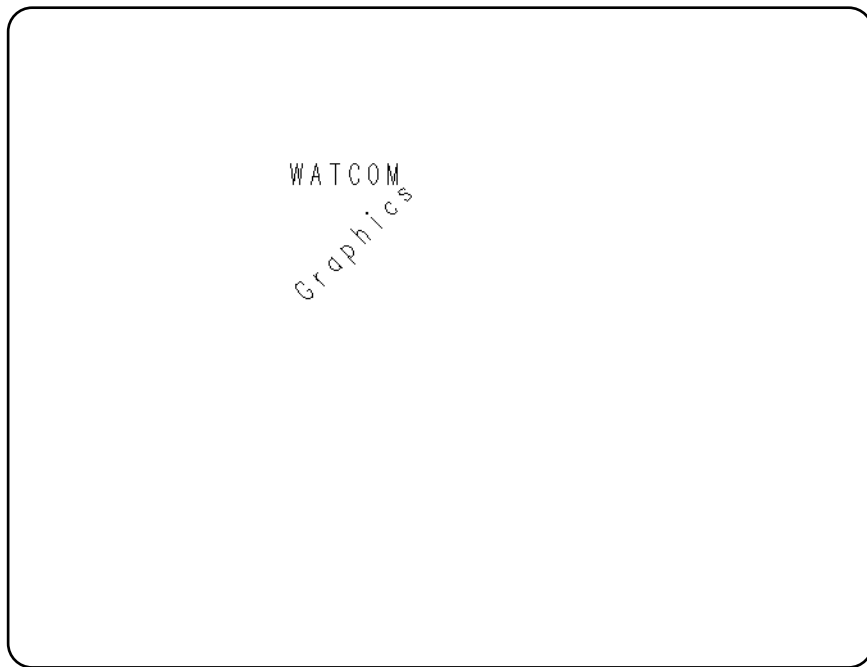
See Also: _grtext, _gettextsettings

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _grtext( 200, 100, 'WATCOM'c )
call _settextorient( 1, 1 )
call _grtext( 200, 200, 'Graphics'c )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

_settextpath

Synopsis: subroutine _settextpath(path)
 integer*2 path

Description: The _settextpath routine sets the current text path to have the value of the *path* argument. The text path specifies the writing direction of the text displayed by the _grtext routine. The argument can have one of the following values:

_PATH_RIGHT subsequent characters are drawn to the right of the previous character

_PATH_LEFT subsequent characters are drawn to the left of the previous character

_PATH_UP subsequent characters are drawn above the previous character

_PATH_DOWN subsequent characters are drawn below the previous character

The default value of the text path is *_PATH_RIGHT*.

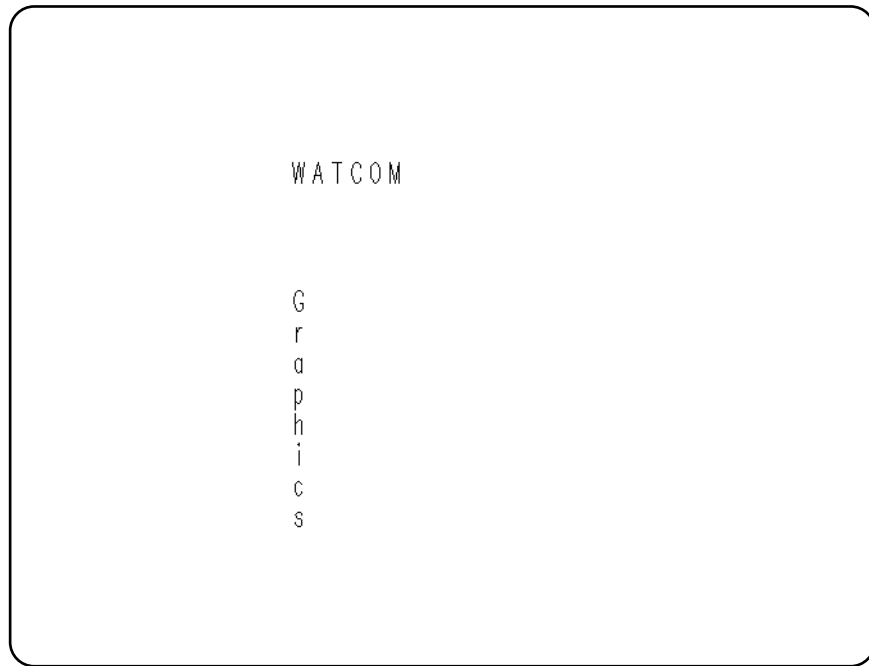
See Also: _grtext, _gettextsettings

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _grtext( 200, 100, 'WATCOM'c )
call _settextpath( _PATH_DOWN )
call _grtext( 200, 200, 'Graphics'c )
pause
call _setvideomode( _DEFAULTMODE )
end
```

produces the following:



Classification: PC Graphics

Systems: DOS

_settextposition

Synopsis: record /rccoord/ function _settextposition(row, col)
 integer*2 row, col

Description: The _settextposition routine sets the current output position for text to be (row,col) where this position is in terms of characters, not pixels.

The text position is relative to the current text window. It defaults to the top left corner of the screen, (1,1), when a new video mode is selected, or when a new text window is set. The position is updated as text is drawn with the _outtext and _outmem routines.

Note that the output position for graphics output differs from that for text output. The output position for graphics output can be set by use of the _moveto routine.

Returns: The _settextposition routine returns, as an rccoord structure, the previous output position for text.

See Also: _gettextposition, _outtext, _outmem, _settextwindow, _moveto

Example:

```
include 'graphapi.fi'
include 'graph.fi'

record /rccoord/ old_pos

call _setvideomode( _TEXTC80 )
old_pos = _gettextposition()
call _settextposition( 10, 40 )
call _outtext( 'WATCOM Graphics'c )
call _settextposition( old_pos.row, old_pos.col )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _settextrows(rows)
 integer*2 rows

Description: The _settextrows routine selects the number of rows of text displayed on the screen. The number of rows is specified by the argument *rows*. Computers equipped with EGA, MCGA and VGA adapters can support different numbers of text rows. The number of rows that can be selected depends on the current video mode and the type of monitor attached.

If the argument *rows* has the value *_MAXTEXTROWS*, the maximum number of text rows will be selected for the current video mode and hardware configuration. In text modes the maximum number of rows is 43 for EGA adapters, and 50 for MCGA and VGA adapters. Some graphics modes will support 43 rows for EGA adapters and 60 rows for MCGA and VGA adapters.

Returns: The _settextrows routine returns the number of screen rows when the number of rows is set successfully; otherwise, zero is returned.

See Also: _getvideoconfig, _setvideomode, _setvideomoderows

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer valid_rows(8)/
+      14, 25, 28, 30,
+      34, 43, 50, 60/

integer i, j, rows
character*80 buff

do i = 0, 7
  rows = valid_rows( i )
  if( _settextrows( rows ) .eq. rows )then
    do j = 1, rows
      write( buff, '( "Line ", i2, a1 )' )
+      j, char(0)
      call _setttextposition( j, 1 )
      call _outtext( buff )
    enddo
    pause
  endif
enddo
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

_settextrows

Synopsis: subroutine _settextwindow(row1, col1, row2, col2)
 integer*2 row1, col1
 integer*2 row2, col2

Description: The _settextwindow routine sets the text window to be the rectangle with a top left corner at (row1,col1) and a bottom right corner at (row2,col2) . These coordinates are in terms of characters not pixels.

The initial text output position is (1,1) . Subsequent text positions are reported (by the _gettextposition routine) and set (by the _outtext,_outmem and _settextposition routines) relative to this rectangle.

Text is displayed from the current output position for text proceeding along the current row and then downwards. When the window is full, the lines scroll upwards one line and then text is displayed on the last line of the window.

See Also: _gettextposition,_outtext,_outmem,_settextposition

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i
integer*2 r1, c1, r2, c2
character*80 buff

call _setvideomode( _TEXTC80 )
call _gettextwindow( r1, c1, r2, c2 )
call _settextwindow( 5, 20, 20, 40 )
do i = 1, 20
    write( buff, '( "Line ", i2, a1, a1 ) ' )
+      i, char(10), char(0)
    call _outtext( buff )
enddo
pause
call _settextwindow( r1, c1, r2, c2 )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _setvideomode(mode)
 integer*2 mode

Description: The _setvideomode routine sets the video mode according to the value of the *mode* argument. The value of *mode* can be one of the following:

Mode	Type	Size	Colors	Adapter
_MAXRESMODE	(graphics mode with highest resolution)			
_MAXCOLORMODE	(graphics mode with most colors)			
_DEFAULTMODE	(restores screen to original mode)			
_TEXTBW40	M, T	40 x 25	16	MDPA, HGC, VGA, SVGA
_TEXTC40	C, T	40 x 25	16	CGA, EGA, MCGA, VGA, SVGA
_TEXTBW80	M, T	80 x 25	16	MDPA, HGC, VGA, SVGA
_TEXTC80	C, T	80 x 25	16	CGA, EGA, MCGA, VGA, SVGA
_MRES4COLOR	C, G	320 x 200	4	CGA, EGA, MCGA, VGA, SVGA
_MRESNOCOLOR	C, G	320 x 200	4	CGA, EGA, MCGA, VGA, SVGA
_HRESBW	C, G	640 x 200	2	CGA, EGA, MCGA, VGA, SVGA
_TEXTMONO	M, T	80 x 25	16	MDPA, HGC, VGA, SVGA
_HERCMONO	M, G	720 x 350	2	HGC
_MRES16COLOR	C, G	320 x 200	16	EGA, VGA, SVGA
_HRES16COLOR	C, G	640 x 200	16	EGA, VGA, SVGA
_ERESNOCOLOR	M, G	640 x 350	4	EGA, VGA, SVGA
_ERESCOLOR	C, G	640 x 350	4/16	EGA, VGA, SVGA
_VRES2COLOR	C, G	640 x 480	2	MCGA, VGA, SVGA
_VRES16COLOR	C, G	640 x 480	16	VGA, SVGA
_MRES256COLOR	C, G	320 x 200	256	MCGA, VGA, SVGA
_URES256COLOR	C, G	640 x 400	256	SVGA
_VRES256COLOR	C, G	640 x 480	256	SVGA
_SVRES16COLOR	C, G	800 x 600	16	SVGA
_SVRES256COLOR	C, G	800 x 600	256	SVGA
_XRES16COLOR	C, G	1024 x 768	16	SVGA
_XRES256COLOR	C, G	1024 x 768	256	SVGA

In the preceding table, the Type column contains the following letters:

M	indicates monochrome; multiple colors are shades of grey
C	indicates color
G	indicates graphics mode; size is in pixels
T	indicates text mode; size is in columns and rows of characters

The Adapter column contains the following codes:

MDPA	IBM Monochrome Display/Printer Adapter
CGA	IBM Color Graphics Adapter
EGA	IBM Enhanced Graphics Adapter
VGA	IBM Video Graphics Array
MCGA	IBM Multi-Color Graphics Array
HGC	Hercules Graphics Adapter
SVGA	SuperVGA adapters

The modes `_MAXRESMODE` and `_MAXCOLORMODE` will select from among the video modes supported by the current graphics adapter the one that has the highest resolution or the greatest number of colors. The video mode will be selected from the standard modes, not including the SuperVGA modes.

Selecting a new video mode resets the current output positions for graphics and text to be the top left corner of the screen. The background color is reset to black and the default color value is set to be one less than the number of colors in the selected mode.

Returns: The `_setvideomode` routine returns the number of text rows when the new mode is successfully selected; otherwise, zero is returned.

See Also: `_getvideoconfig`, `_settextrows`, `_setvideomoderows`

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer mode
record /videoconfig/ vc
character*80 buff

call _getvideoconfig( vc )
select( vc.adapter )
case( _VGA, _SVGA )
    mode = _VRES16COLOR
case( _MCGA )
    mode = _MRES256COLOR
case( _EGA )
    if( vc.monitor .eq. _MONO )then
        mode = _ERESNOCOLOR
    else
        mode = _ERESCOLOR
    endif
case( _CGA )
    mode = _MRES4COLOR
case( _HERCULES )
    mode = _HERCMONO
case default
    stop 'No graphics adapter'
endselect
if( _setvideomode( mode ) .ne. 0 )then
    call _getvideoconfig( vc )
    write( buff,
+         '( i3, '' x '', i3, '' x '', i3, a1 )' )
+         vc.numxpixels, vc.numypixels,
+         vc.numcolors, char(0)
    call _outtext( buff )
    pause
    call _setvideomode( _DEFAULTMODE )
endif
end
```

Classification: PC Graphics

Systems: DOS

_setvideomode

Synopsis: integer*2 function _setvideomoderows(mode, rows)
 integer*2 mode
 integer*2 rows

Description: The _setvideomoderows routine selects a video mode and the number of rows of text displayed on the screen. The video mode is specified by the argument *mode* and is selected with the _setvideomode routine. The number of rows is specified by the argument *rows* and is selected with the _settextrows routine.

Computers equipped with EGA, MCGA and VGA adapters can support different numbers of text rows. The number of rows that can be selected depends on the video mode and the type of monitor attached.

Returns: The _setvideomoderows routine returns the number of screen rows when the mode and number of rows are set successfully; otherwise, zero is returned.

See Also: _getvideoconfig, _setvideomode, _settextrows

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer rows
character*80 buff

rows = _setvideomoderows( _TEXTC80, _MAXTEXTROWS )
if( rows .ne. 0 )then
    write( buff,
+         ' (''Number of rows is '', i2, a1 )' )
+         rows, char(0)
    call _outtext( buff )
    pause
    call _setvideomode( _DEFAULTMODE )
endif
end
```

Classification: PC Graphics

Systems: DOS

_setvieworg

Synopsis: record /xycoord/ function _setvieworg(x, y)
 integer*2 x, y

Description: The _setvieworg routine sets the origin of the view coordinate system, (0,0), to be located at the physical point (x,y). This causes subsequently drawn images to be translated by the amount (x,y).

Returns: The _setvieworg routine returns, as an xycoord structure, the physical coordinates of the previous origin.

See Also: _getviewcoord, _getphyscoord, _setcliprgn, _setviewport

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _VRES16COLOR )
call _setvieworg( 320, 240 )
call _ellipse( _GBORDER, -200, -150, 200, 150 )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: subroutine _setviewport(x1, y1, x2, y2)
 integer*2 x1, y1
 integer*2 x2, y2

Description: The _setviewport routine restricts the display of graphics output to the clipping region and then sets the origin of the view coordinate system to be the top left corner of the region. This region is a rectangle whose opposite corners are established by the physical points (x1,y1) and (x2,y2) .

The _setviewport routine does not affect text output using the _outtext and _outmem routines. To control the location of text output, see the _settextwindow routine.

See Also: _setcliprgn, _setvieworg, _settextwindow, _setwindow

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer XSIZE, YSIZE
parameter (XSIZE=380)
parameter (YSIZE=280)

call _setvideomode( _VRES16COLOR )
call _setviewport( 130, 100,
+               130 + XSIZE, 100 + YSIZE )
call _ellipse( _GBORDER, 0, 0, XSIZE, YSIZE )
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _setvisualpage(pagenum)
 integer*2 pagenum

Description: The _setvisualpage routine selects the page (in memory) from which graphics output is displayed. The page to be selected is given by the *pagenum* argument.

Only some combinations of video modes and hardware allow multiple pages of graphics to exist. When multiple pages are supported, the active page may differ from the visual page. The graphics information in the visual page determines what is displayed upon the screen. Animation may be accomplished by alternating the visual page. A graphics page can be constructed without affecting the screen by setting the active page to be different than the visual page.

The number of available video pages can be determined by using the _getvideoconfig routine. The default video page is 0.

Returns: The _setvisualpage routine returns the number of the previous page when the visual page is set successfully; otherwise, a negative number is returned.

See Also: _getvisualpage, _setactivepage, _getactivepage, _getvideoconfig

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer old_apage, old_vpage

call _setvideomode( _HRES16COLOR )
old_apage = _getactivepage()
old_vpage = _getvisualpage()
! draw an ellipse on page 0
call _setactivepage( 0 )
call _setvisualpage( 0 )
call _ellipse( _GFillInterior, 100, 50,
+              540, 150 )
! draw a rectangle on page 1
call _setactivepage( 1 )
call _rectangle( _GFillInterior, 100, 50,
+               540, 150 )
pause
! display page 1
call _setvisualpage( 1 )
pause
call _setactivepage( old_apage )
call _setvisualpage( old_vpage )
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _setwindow(invert, x1, y1, x2, y2)
 logical invert
 double precision x1, y1
 double precision x2, y2

Description: The _setwindow routine defines a window for the window coordinate system. Window coordinates are specified as a user-defined range of values. This allows for consistent pictures regardless of the video mode.

The window is defined as the region with opposite corners established by the points (x1,y1) and (x2,y2) . The argument *invert* specifies the direction of the y-axis. If the value is .TRUE., the y values increase from the bottom of the screen to the top, otherwise, the y values increase as you move down the screen.

The window defined by the _setwindow routine is displayed in the current viewport. A viewport is defined by the _setviewport routine.

By default, the window coordinate system is defined with the point (0.0,0.0) located at the lower left corner of the screen, and the point (1.0,1.0) at the upper right corner.

Returns: The _setwindow routine returns a non-zero value when the window is set successfully; otherwise, zero is returned.

See Also: _setviewport

Example:

```
include 'graphapi.fi'
include 'graph.fi'

call _setvideomode( _MAXRESMODE )
call draw_house( 'Default window'c )
call _setwindow( .TRUE., -0.5, -0.5, 1.5, 1.5 )
call draw_house( 'Larger window'c )
call _setwindow( .TRUE., 0.0, 0.0, 0.5, 1.0 )
call draw_house( 'Left side'c )
call _setvideomode( _DEFAULTMODE )
end

subroutine draw_house( msg )

include 'graph.fi'
character*80 msg

call _clearscreen( _GCLEARSCREEN )
call _outtext( msg )
call _rectangle_w( _GBORDER, 0.2, 0.1, 0.8, 0.6 )
call _moveto_w( 0.1, 0.5 )
call _lineto_w( 0.5, 0.9 )
call _lineto_w( 0.9, 0.5 )
call _arc_w( 0.4, 0.5, 0.6, 0.3,
+           0.6, 0.4, 0.4, 0.4 )
call _rectangle_w( _GBORDER, 0.4, 0.1, 0.6, 0.4 )
pause
end
```

Classification: PC Graphics

Systems: DOS

_unregisterfonts

Synopsis: subroutine _unregisterfonts()

Description: The _unregisterfonts routine frees the memory previously allocated by the _registerfonts routine. The currently selected font is also unloaded.

Attempting to use the _setfont routine after calling _unregisterfonts will result in an error.

See Also: _registerfonts, _setfont, _getfontinfo, _outgtext, _getgtextextent, _setgtextvector, _getgtextvector

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i, n
character*10 buff

call _setvideomode( _VRES16COLOR )
n = _registerfonts( '*.fon'c )
do i = 0, n - 1
    write( buff, '( ''n'', i2.2, a1 )' ) i, char(0)
    call _setfont( buff )
    call _moveto( 100, 100 )
    call _outgtext( 'WATCOM Graphics'c )
    pause
    call _clearscreen( _GCLEARSCREEN )
enddo
call _unregisterfonts()
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

Synopsis: integer*2 function _wrapon(wrap)
 integer*2 wrap

Description: The _wrapon routine is used to control the display of text when the text output reaches the right side of the text window. This is text displayed with the _outtext and _outmem routines. The *wrap* argument can take one of the following values:

_GWRAPON causes lines to wrap at the window border

_GWRAPOFF causes lines to be truncated at the window border

Returns: The _wrapon routine returns the previous setting for wrapping.

See Also: _outtext, _outmem, _settextwindow

Example:

```
include 'graphapi.fi'
include 'graph.fi'

integer i
character buff*80

call _setvideomode( _TEXTC80 )
call _settextwindow( 5, 20, 20, 30 )
call _wrapon( _GWRAPOFF )
do i = 1, 3
    call _settextposition( 2 * i, 1 )
    write( buff,
+         ' (''Very very long line '', i2, a1)' )
+         i, char(0)
    call _outtext( buff )
enddo
call _wrapon( _GWRAPON )
do i = 4, 6
    call _settextposition( 2 * i, 1 )
    write( buff,
+         ' (''Very very long line '', i2, a1)' )
+         i, char(0)
    call _outtext( buff )
enddo
pause
call _setvideomode( _DEFAULTMODE )
end
```

Classification: PC Graphics

Systems: DOS

A

_ANALOGCOLOR 47
 _ANALOGMONO 47
 _arc 10, 4, 20, 25, 108
 _arc_w 10
 _arc_wxy 10

B

_BASE 146
 _BOTTOM 146

C

_CAP 146
 _CENTER 146
 CGA 160
 _CGA 46
 classes of routines 2
 _clearscreen 13, 122
 _COLOR 47
 coordinate systems 3

D

DEFAULTMODE 160
 _displaycursor 14

E

EGA 160
 _EGA 46
 _ellipse 15, 4, 26
 _ellipse_w 15
 _ellipse_wxy 15

_ENHANCED 47
 ERESCOLOR 160
 ERESNOCOLOR 160

F

_floodfill 17, 26
 _floodfill_w 17

G

_GAND 36, 113, 144
 _GBORDER 15, 109, 111, 115
 _GCLEARSCREEN 13
 _GCURSOROFF 14
 _GCURSORON 14
 _getactivepage 18
 _getarcinfo 20
 _getbkcolor 22
 _getcliprgn 23
 _getcolor 24
 _getcurrentposition 25
 _getcurrentposition_w 25
 _getfillmask 26
 _getfontinfo 27
 _getgettextent 29
 _getgettextvector 30
 _getimage 31, 6, 57, 113
 _getimage_w 31
 _getimage_wxy 31
 _getlinestyle 33
 _getphyscoord 34
 _getpixel 35
 _getpixel_w 35
 _getplotaction 36
 _gettextcolor 38
 _gettextcursor 39
 _gettexttextent 40
 _gettextposition 42, 159
 _gettextsettings 43, 129
 _gettextwindow 45
 _getvideoconfig 46, 2, 18, 51, 118, 126, 166
 _getviewcoord 49
 _getviewcoord_w 49
 _getviewcoord_wxy 49
 _getvisualpage 51
 _getwindowcoord 53

`_GFILLINTERIOR` 15, 109, 111, 115
`_GOR` 36, 113, 144
`_GPRESET` 113
`_GPSET` 36, 113, 144
graphic page 2
graphics adapters 1
graphics include files 8
graphics library 1
graphics routines 1
`GRCLIPPED` 54
`GRERROR` 54
`GRFONTFILENOTFOUND` 54
`GRINSUFFICIENTMEMORY` 54
`GRINVALIDFONTFILE` 54
`GRINVALIDPARAMETER` 54
`GRMODENOTSUPPORTED` 54
`GRNOOUTPUT` 54
`GRNOTINPROPERMODE` 54
`GROK` 54
`_grstatus` **54**
`_grtext` **55**, 40, 43, 55, 62, 64, 66, 129, 131, 146, 152, 154
`_grtext_w` **55**
`_GSCROLLDOWN` 122
`_GSCROLLUP` 122
`_GVIEWPORT` 13
`_GWINDOW` 13
`_GWRAPOFF` 171
`_GWRAPON` 171
`_GXOR` 36, 113, 144

H

`_HALF` 146
`HERCMONO` 160
`_HERCULES` 46
`HGC` 160
`HRES16COLOR` 160
`HRESBW` 160

I

`_imagesize` **57**, 31
`_imagesize_w` **57**
`_imagesize_wxy` **57**

L

`_LEFT` 146
`_lineto` **59**, 4, 25, 61
`_lineto_w` **59**

M

`MAXCOLORMODE` 160
`MAXRESMODE` 160
`MCGA` 160
`_MCGA` 46
`MDPA` 160
`_MDPA` 46
`_MONO` 47
`_moveto` **61**, 25, 42, 62, 156
`_moveto_w` **61**
`MRES16COLOR` 160
`MRES256COLOR` 160
`MRES4COLOR` 160
`MRESNOCOLOR` 160

N

`_NODISPLAY` 46
`_NORMAL` 146

O

`_outgtext` **62**, 29-30, 55, 62, 64, 66, 117, 137, 139
`_outmem` **64**, 5, 38, 42, 55, 62, 64, 66, 133, 148, 156, 159, 165, 171
`_outtext` **66**, 5, 38, 42, 55, 62, 64, 66, 133-134, 148, 156, 159, 165, 171

P

_PATH_DOWN 154
 _PATH_LEFT 154
 _PATH_RIGHT 154
 _PATH_UP 154
 PC Graphics classification 9
 _pg_analyzechart 67
 _pg_analyzechartms 67
 _pg_analyzepie 69
 _pg_analyzescatter 71
 _pg_analyzescatterms 71
 _PG_BARCHART 81
 _pg_chart 73, 67
 _pg_chartms 73, 67
 _pg_chartpie 76, 69
 _pg_chartscatter 78, 71
 _pg_chartscatterms 78, 71
 _PG_COLUMNCHART 81
 _pg_defaultchart 81
 _pg_getchardef 83
 _pg_getpalette 85
 _pg_getstyleset 88
 _pg_hlabelchart 90
 _pg_initchart 92, 7
 _PG_LINECHART 81
 PG_NOPERCENT 81
 PG_PERCENT 81
 _PG_PIECHART 81
 PG_PLAINBARS 81
 PG_POINTANDLINE 81
 PG_POINTONLY 81
 _pg_resetpalette 94
 _pg_resetstyleset 97
 _PG_SCATTERCHART 81
 _pg_setchardef 99
 _pg_setpalette 101
 _pg_setstyleset 104
 PG_STACKEDBARS 81
 _pg_vlabelchart 106
 physical coordinates 3
 _pie 108, 4, 20, 26
 _pie_w 108
 _pie_wxy 108
 _polygon 111, 4, 26
 _polygon_w 111
 _polygon_wxy 111
 _putimage 113, 6, 31
 _putimage_w 113

R

_rectangle 115, 4, 26
 _rectangle_w 115
 _rectangle_wxy 115
 _registerfonts 117, 6, 137, 170
 _remapallpalette 118
 _remappalette 120
 _RIGHT 146

S

_scrolltextwindow 122
 _selectpalette 124
 _setactivepage 126
 _setbkcolor 128
 _setcharsize 129, 5
 _setcharsize_w 129
 _setcharspacing 131
 _setcharspacing_w 131
 _setcliprgn 133, 23
 _setcolor 134, 4, 148
 _setfillmask 135, 4
 _setfont 137, 6, 27, 62, 92, 117, 170
 _setgttextvector 139
 _setlinestyle 140, 4
 _setpixel 142
 _setpixel_w 142
 _setplotaction 144, 4
 _settextalign 146, 5
 _settextcolor 148, 5, 64, 66, 134
 _settextcursor 150, 39
 _settextorient 152, 5
 _settextpath 154
 _settextposition 156, 5, 25, 42, 61, 64, 66, 159
 _settextrows 157, 163
 _settextwindow 159, 5, 42, 45, 122, 133, 165
 _setvideomode 160, 2, 92, 163
 _setvideomoderows 163
 _setvieworg 164, 3, 34, 49
 _setviewport 165, 23, 25, 34, 49, 168
 _setvisualpage 166
 _setwindow 168, 3, 49, 53
 SVGA 160
 _SVGA 46
 SVRES16COLOR 160
 SVRES256COLOR 160

T

TEXTBW40 160
TEXTBW80 160
TEXTC40 160
TEXTC80 160
TEXTMONO 160
_TOP 146

U

_UNKNOWN 46
_unregisterfonts **170**
URES256COLOR 160

V

VGA 160
_VGA 46
view coordinates 3
VRES16COLOR 160
VRES256COLOR 160
VRES2COLOR 160

W

window coordinates 3
_wrapon **171**

X

XRES16COLOR 160
XRES256COLOR 160